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(54) **SYSTEMS AND METHODS FOR EFFICIENT
VEHICLE EXTENT ESTIMATION**

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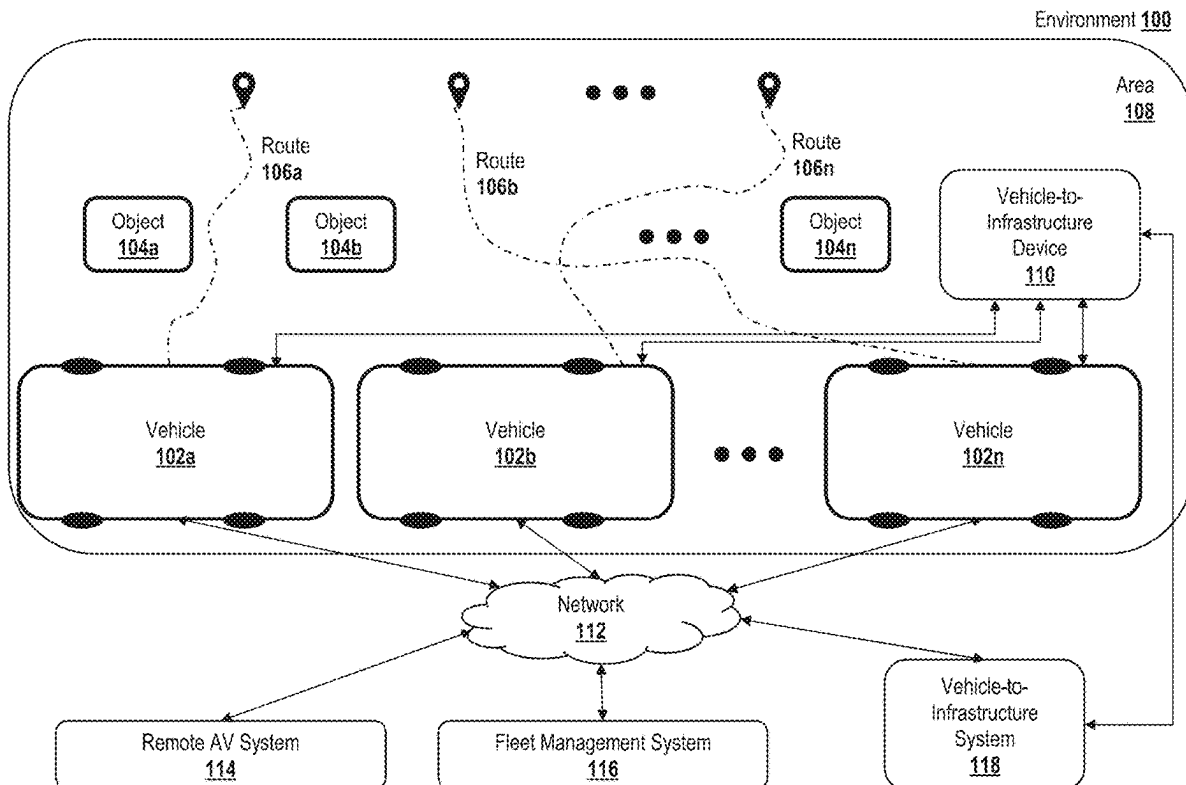
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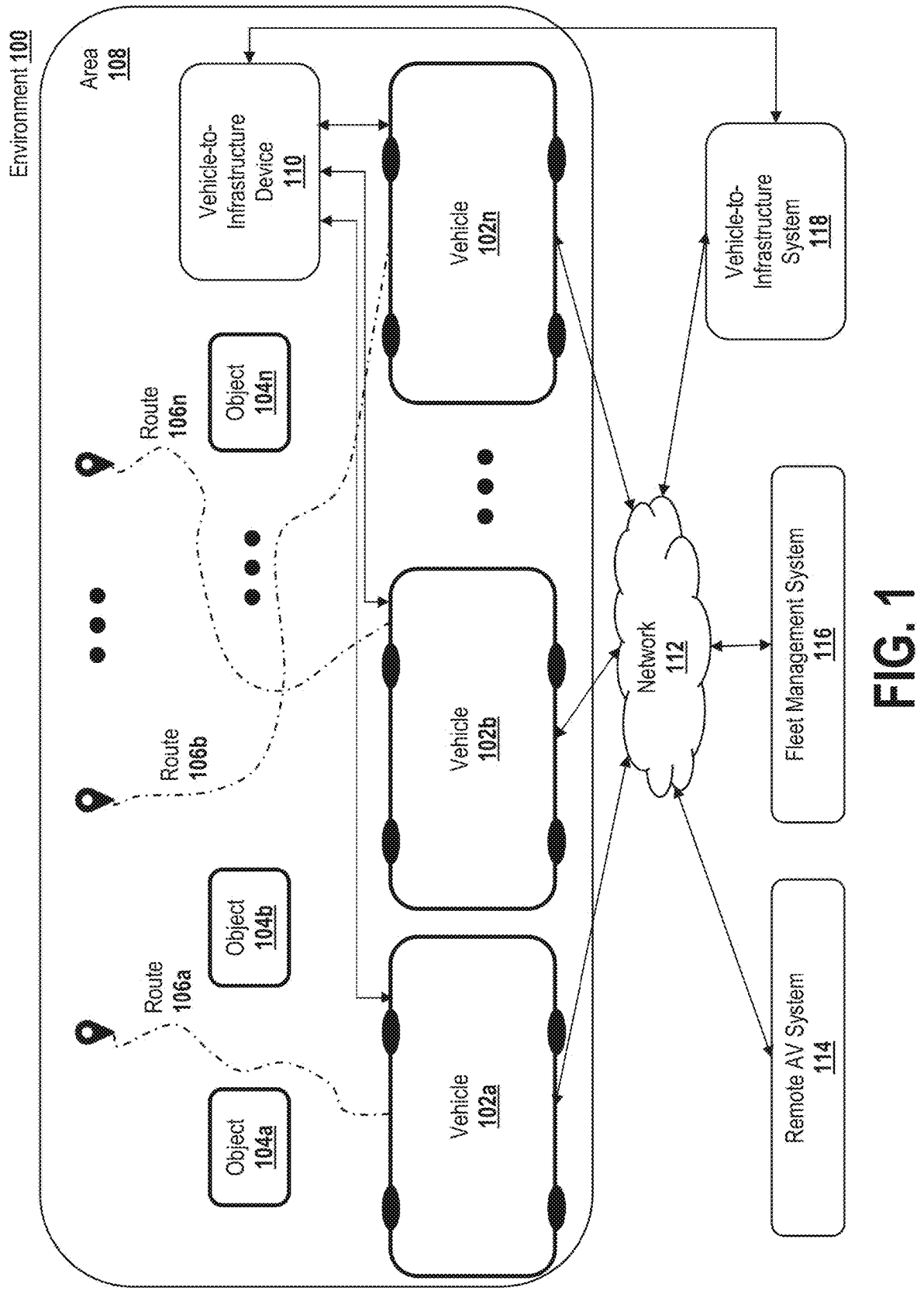
(60) Provisional application No. 63/229,370, filed on Aug.
4, 2021.

(57)

ABSTRACT

Provided are methods for efficient vehicle extent estimation, which can include bounding box generation. Some methods described include determining bounding boxes surrounding detected point clusters according to tangents to convex hulls of the point clusters, and minimizing continuous functions of distances between points and bounding box sides. Accordingly, best-fit bounding boxes are determined more efficiently and quickly, as well as more accurately. Systems and computer program products are also provided.





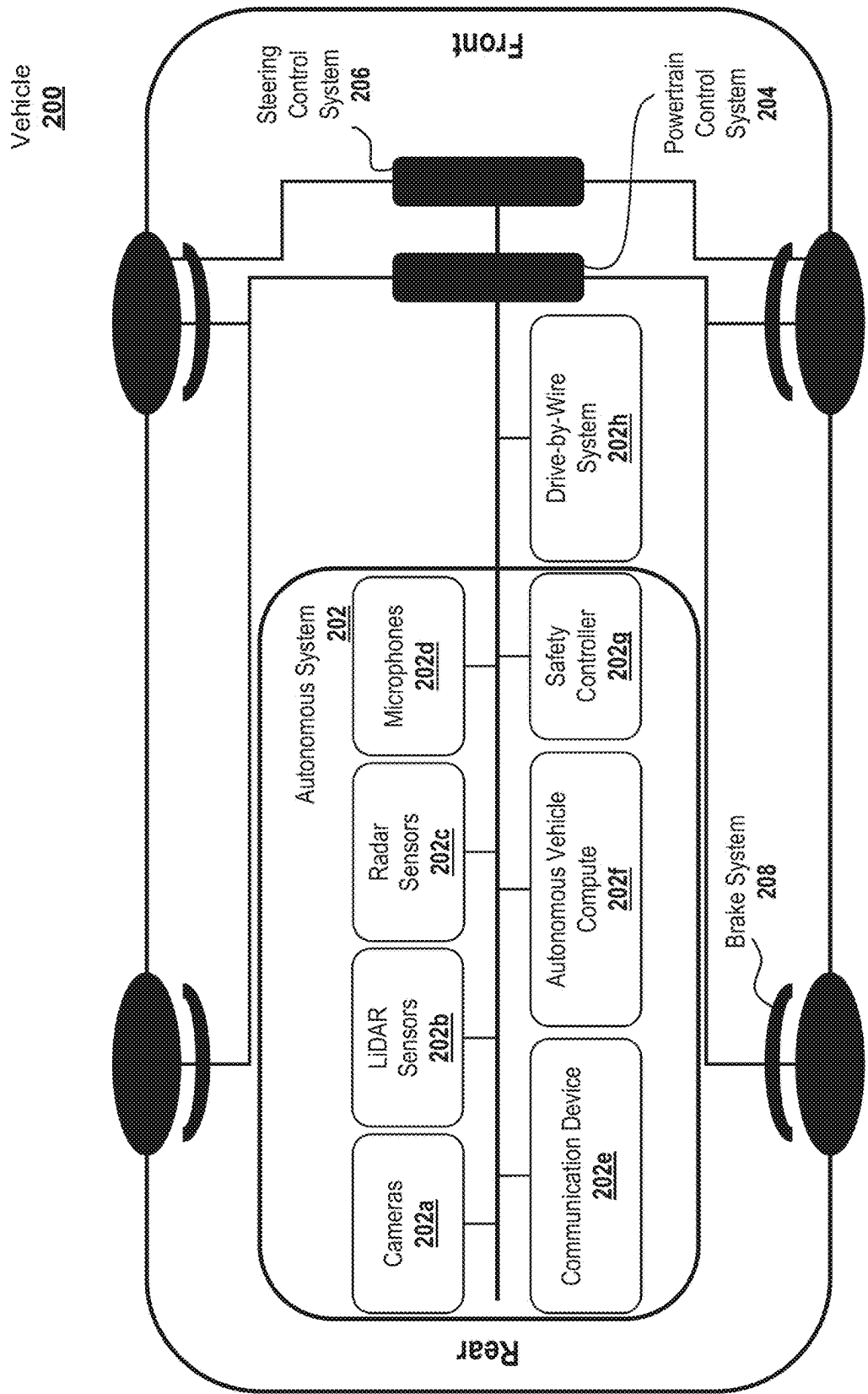


FIG. 2

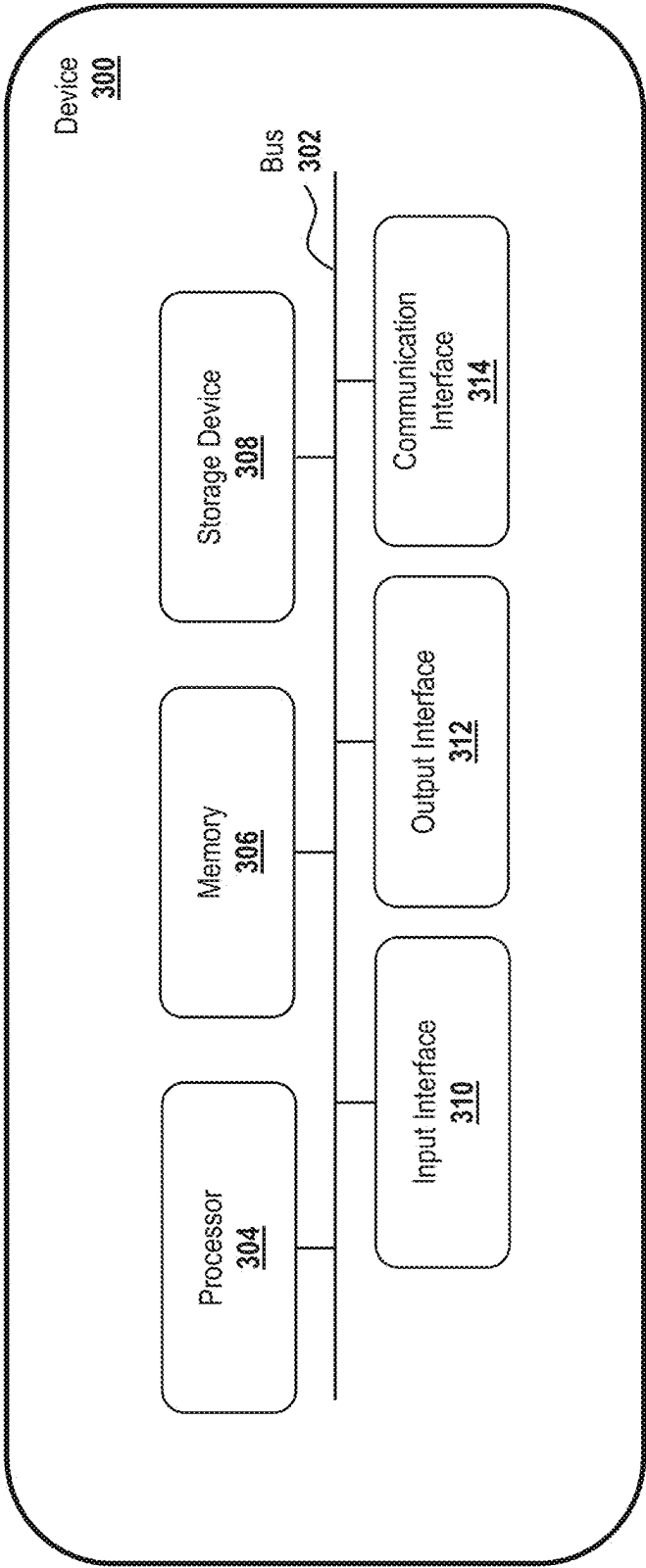


FIG. 3

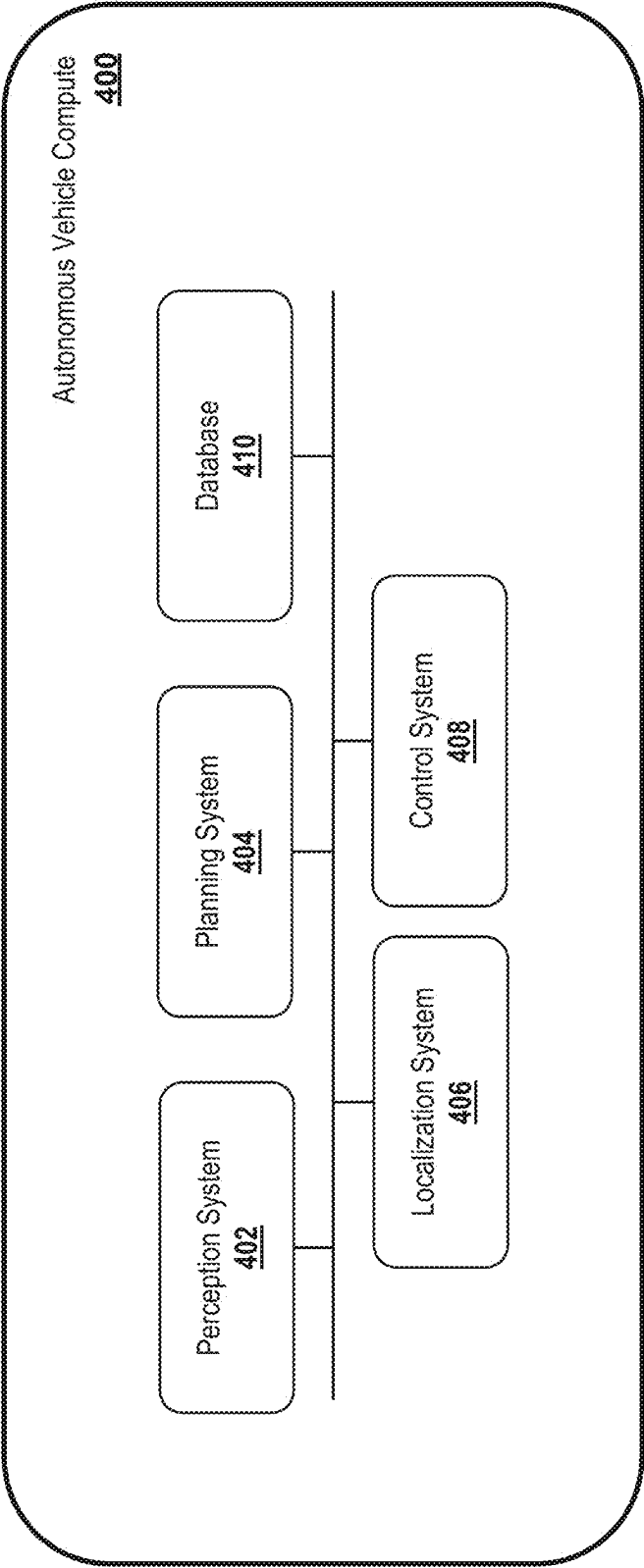


FIG. 4

500

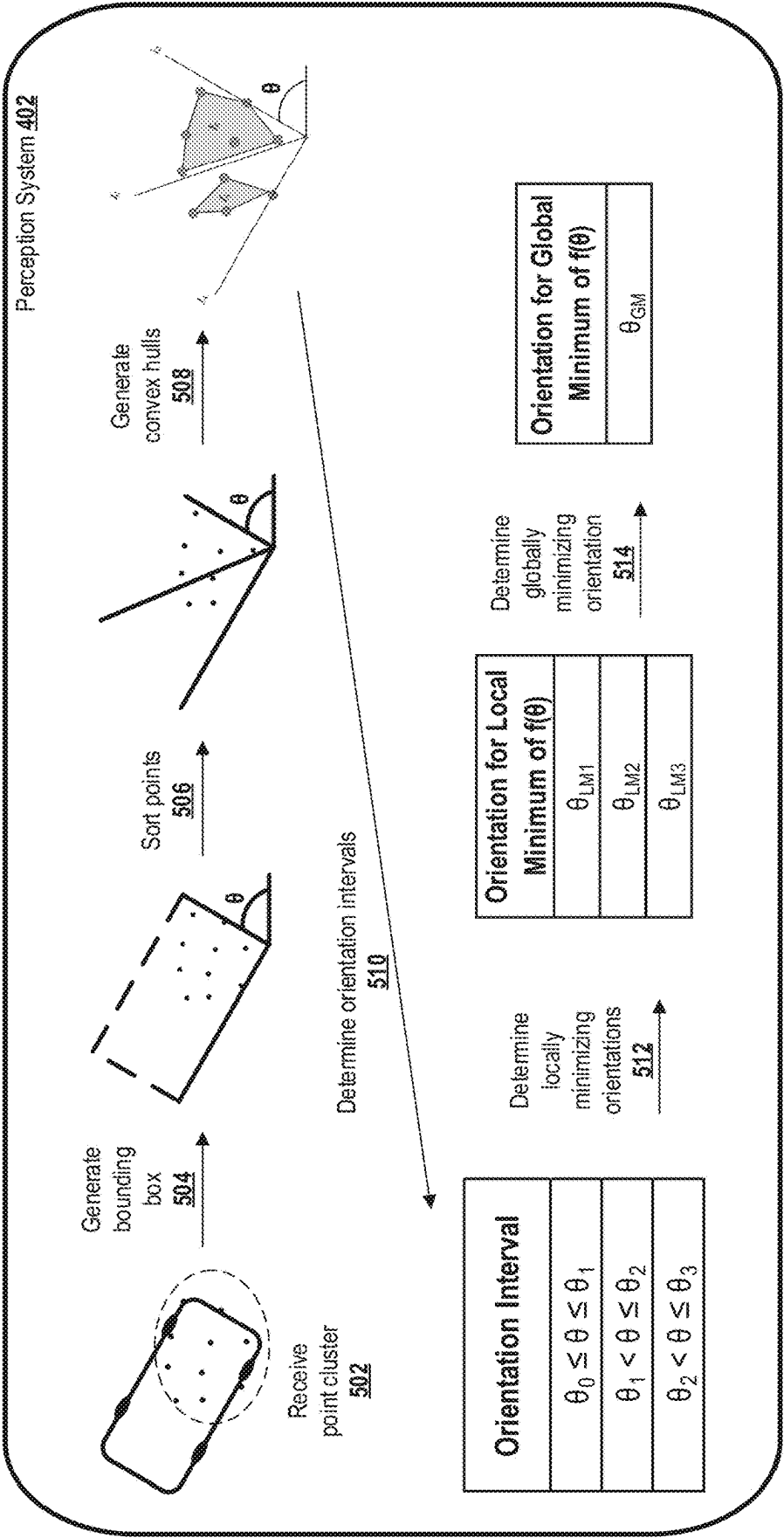


FIG. 5

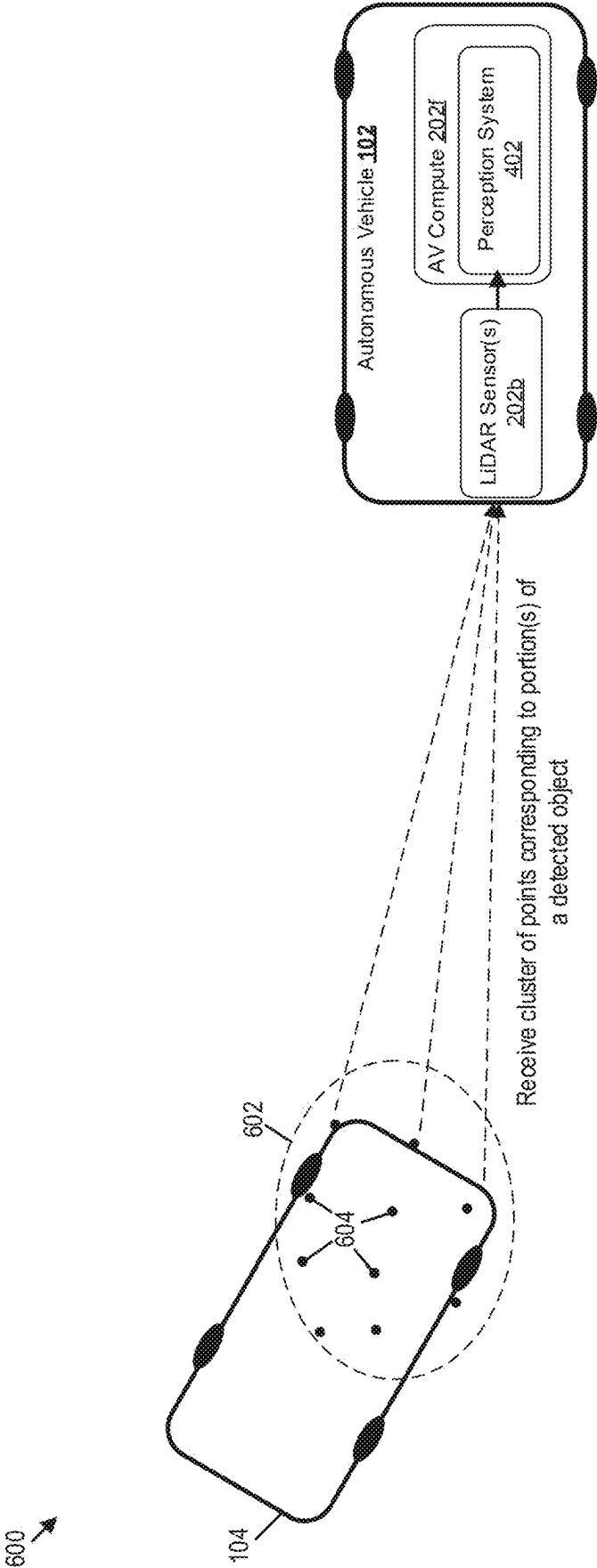


FIG. 6A

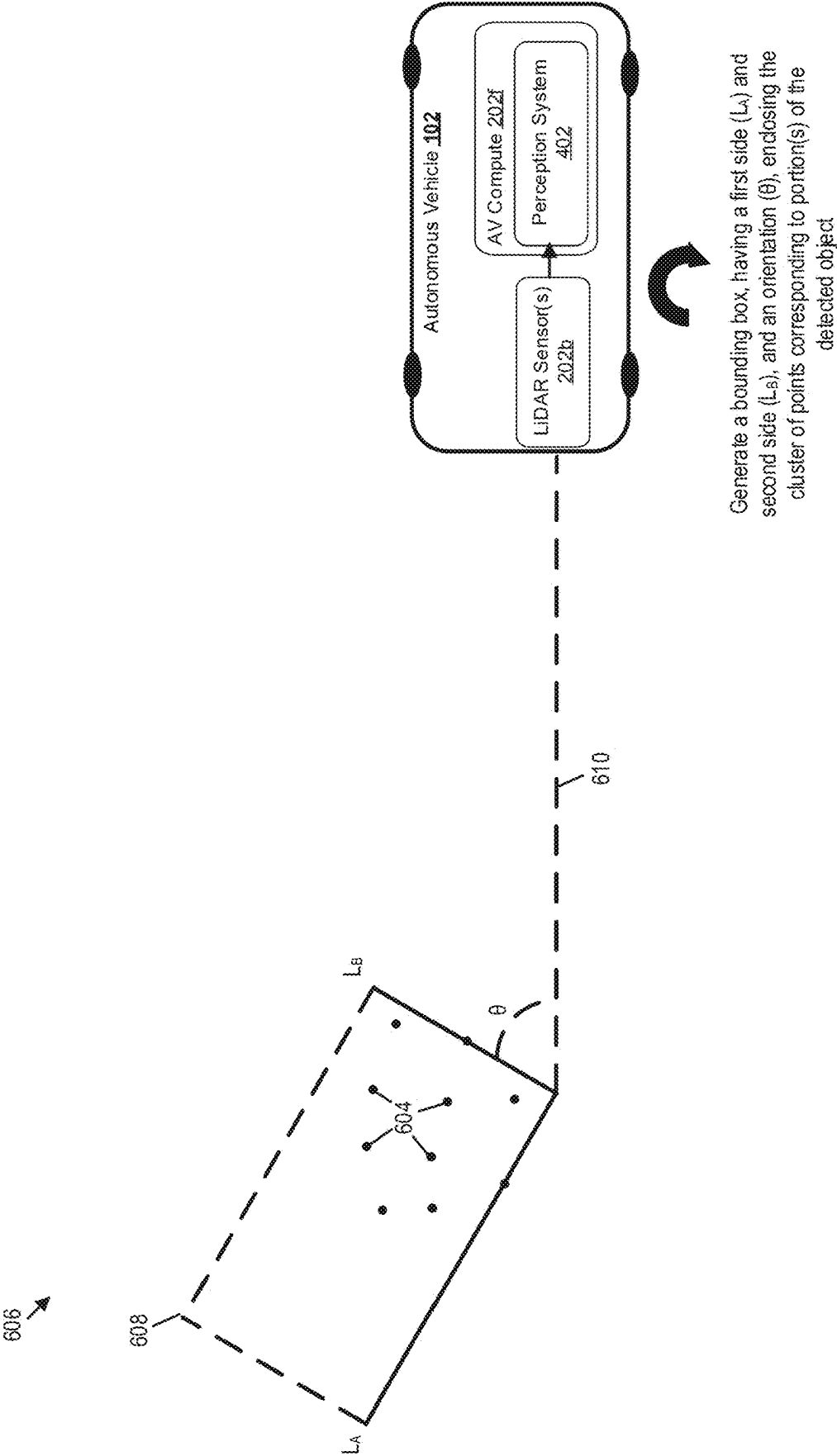


FIG. 6B

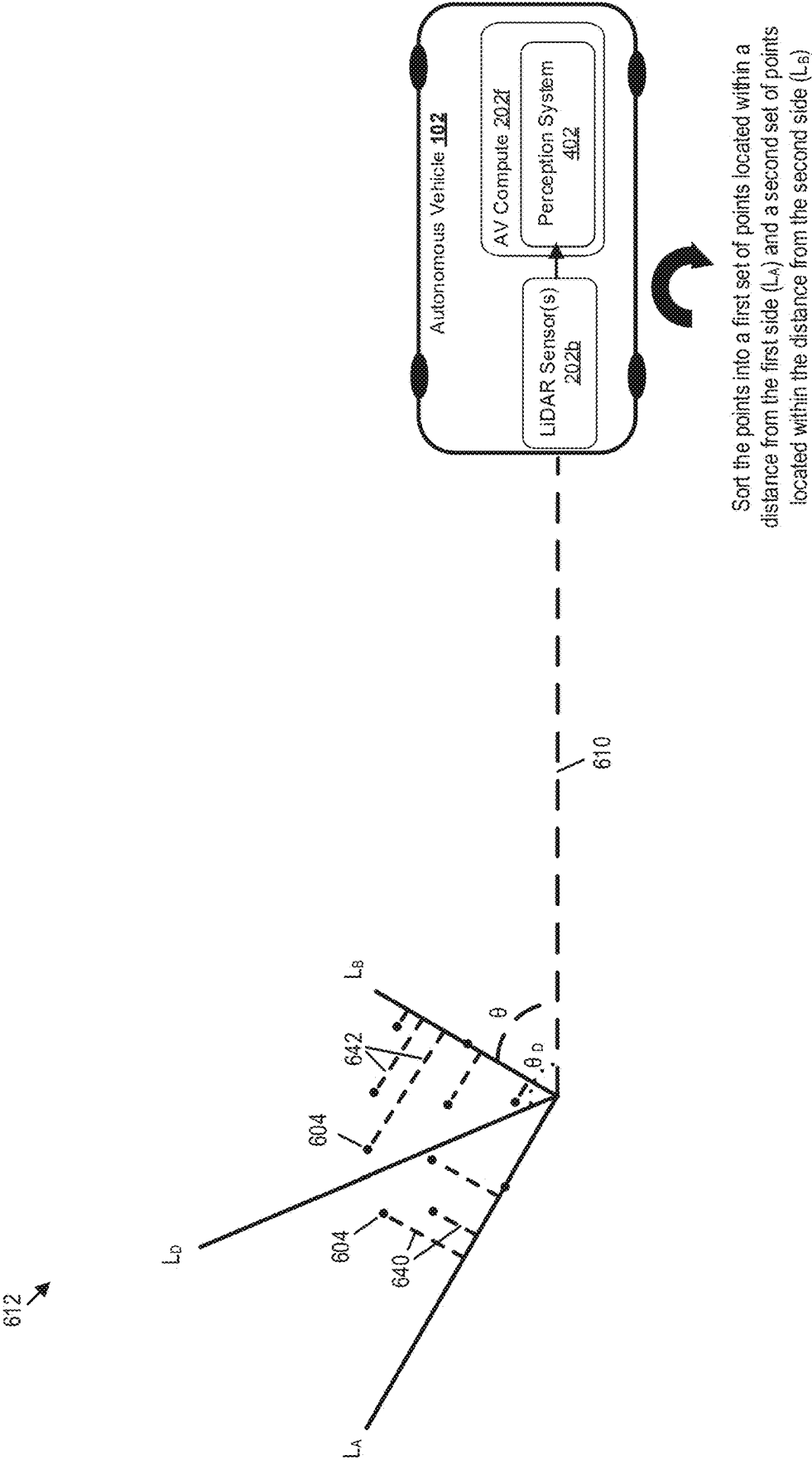
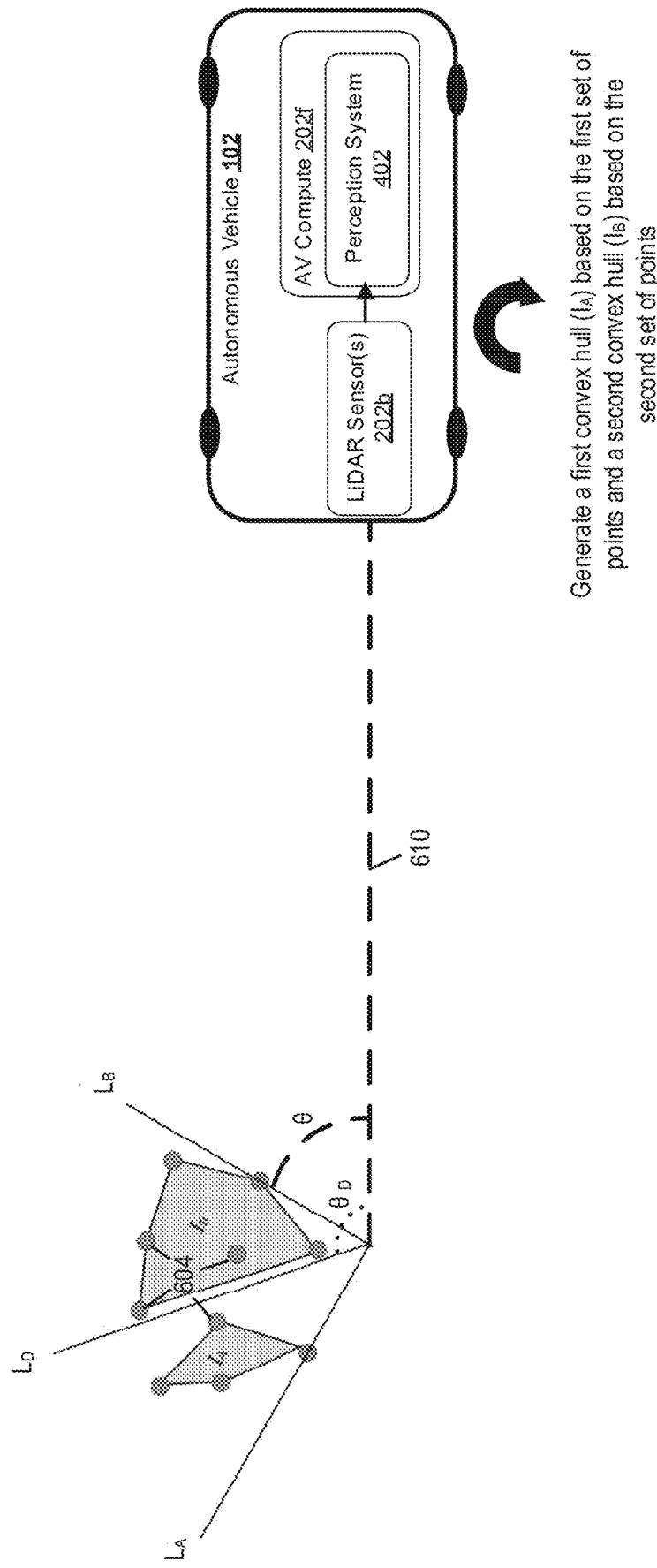


FIG. 6C

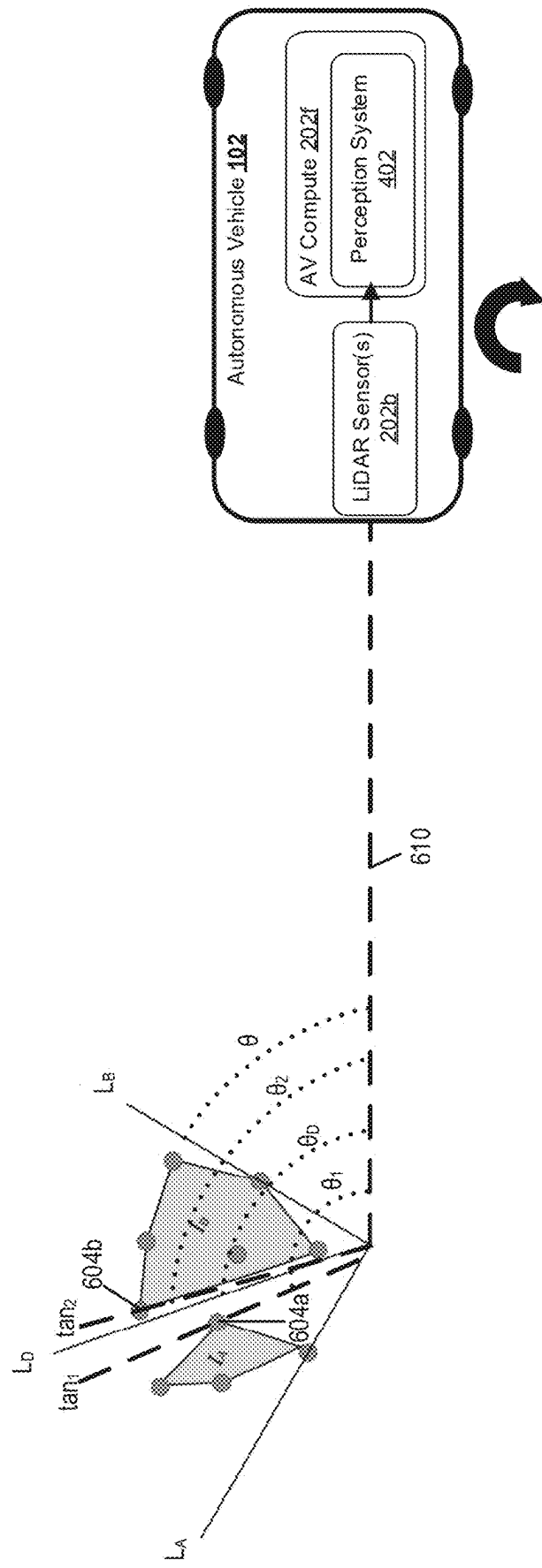
614



Generate a first convex hull (L_A) based on the first set of points and a second convex hull (L_B) based on the second set of points

FIG. 6D

616 →



Determine, based on the first and second convex hulls, intervals of orientations of the bounding box within which the first and second sets of points remain constant

FIG. 6E

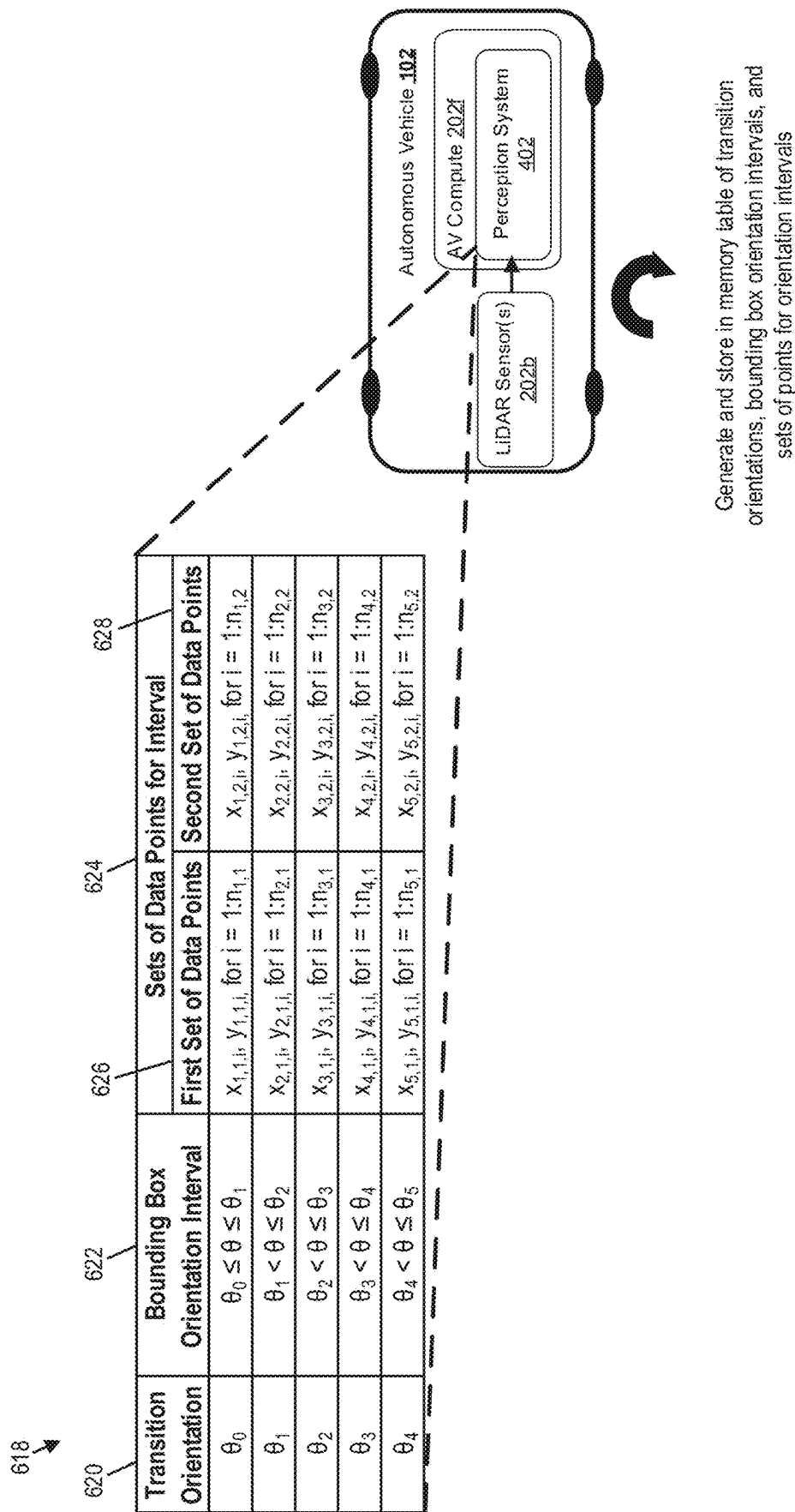


FIG. 6F

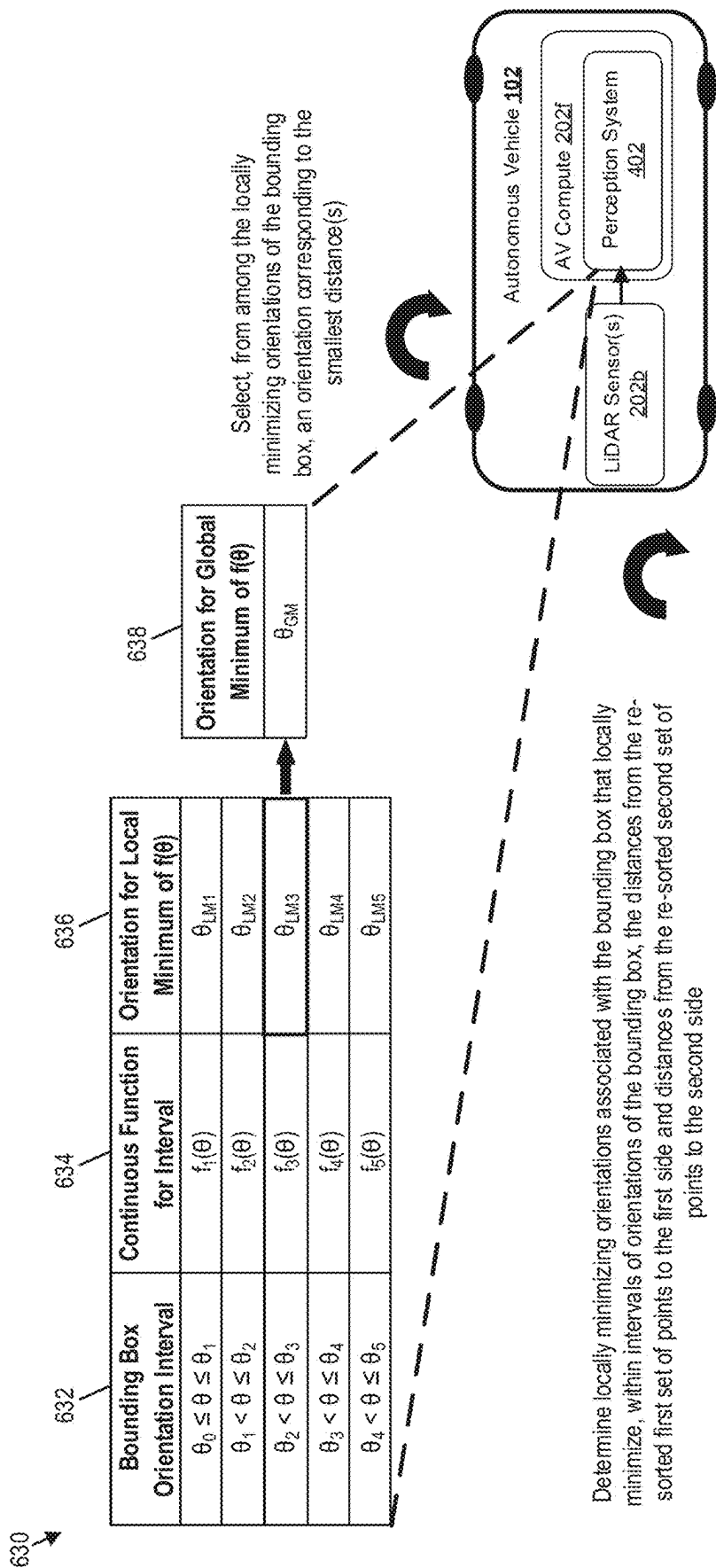
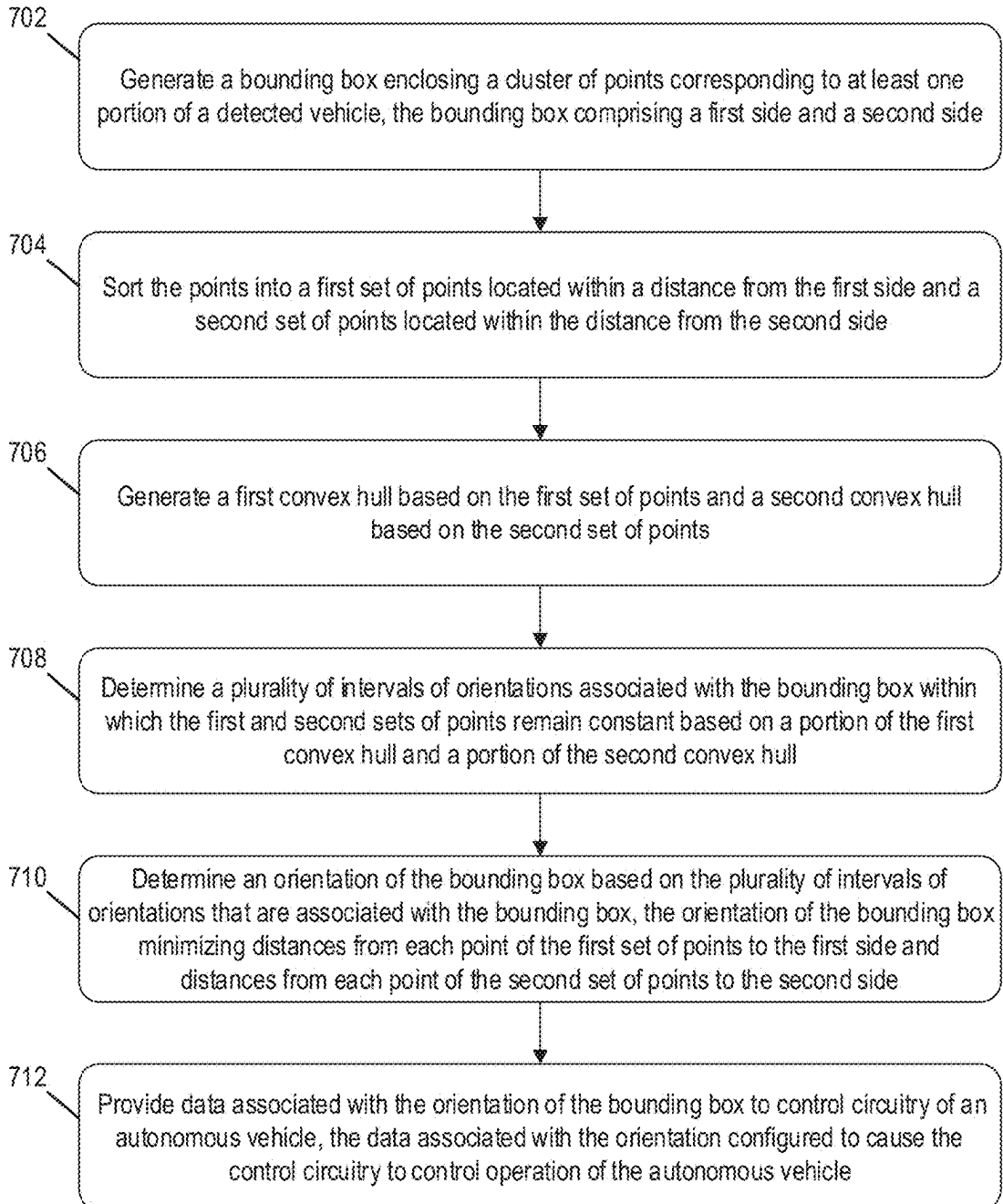


FIG. 6G

700
↓**FIG. 7**

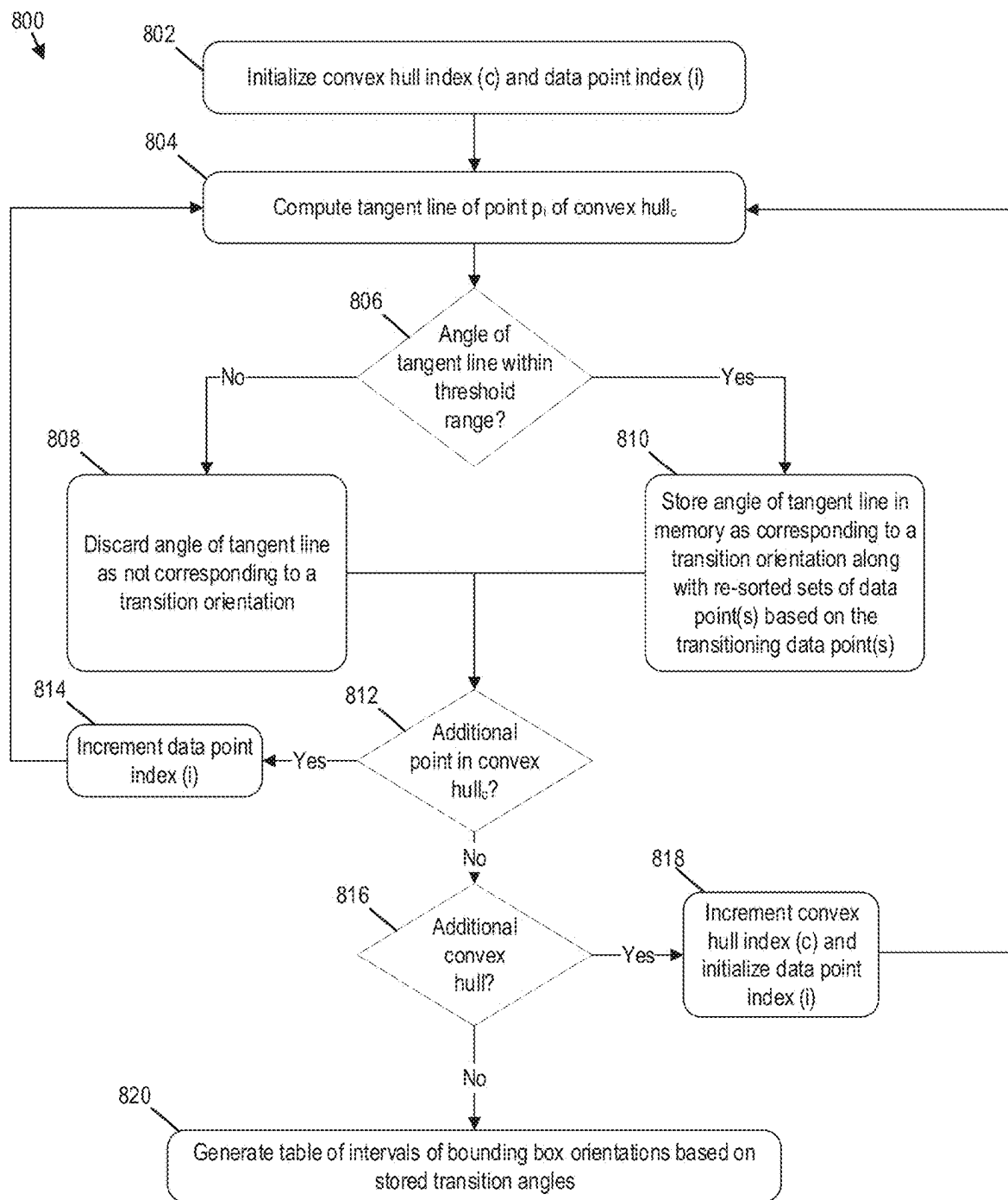


FIG. 8

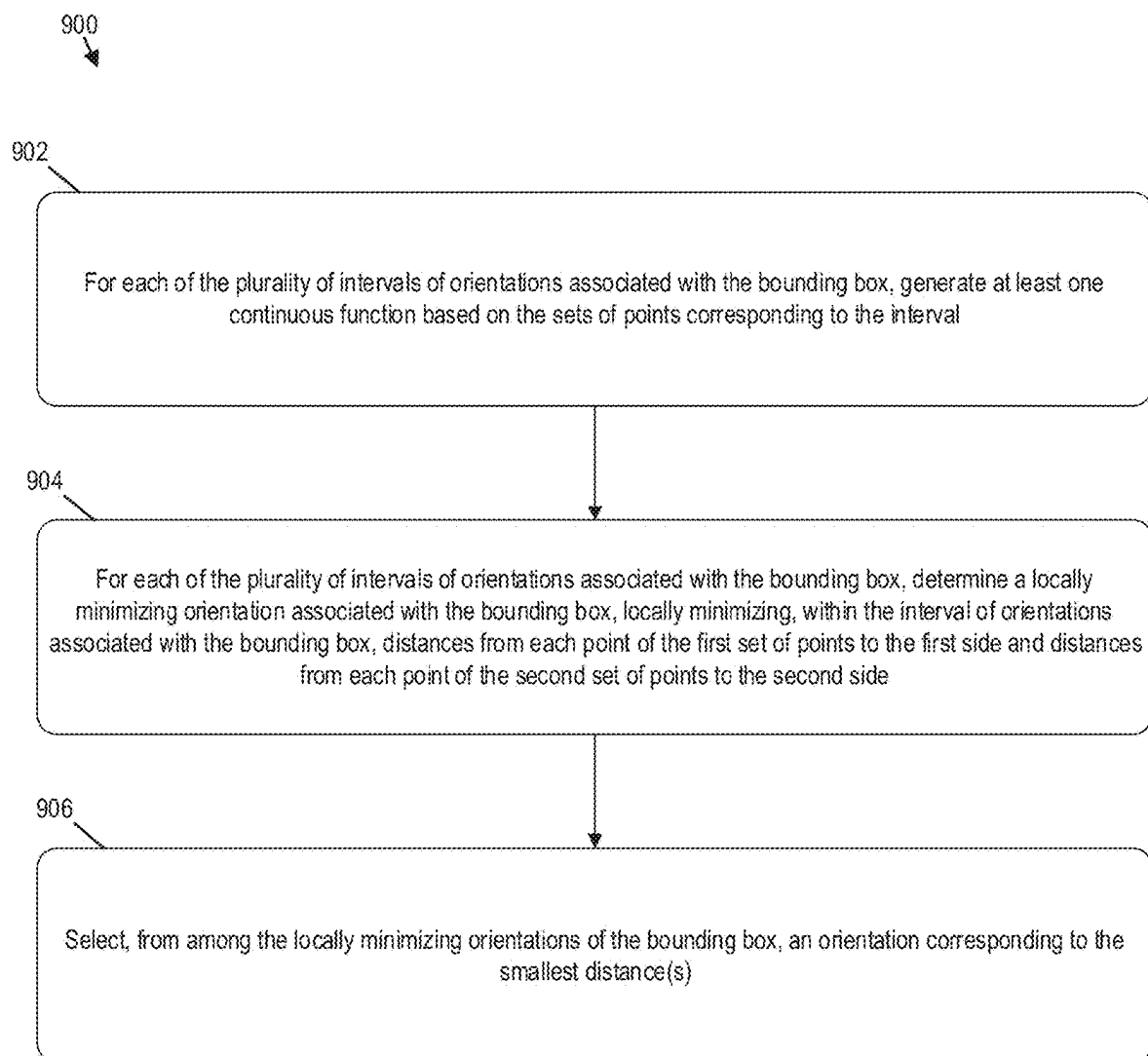


FIG. 9

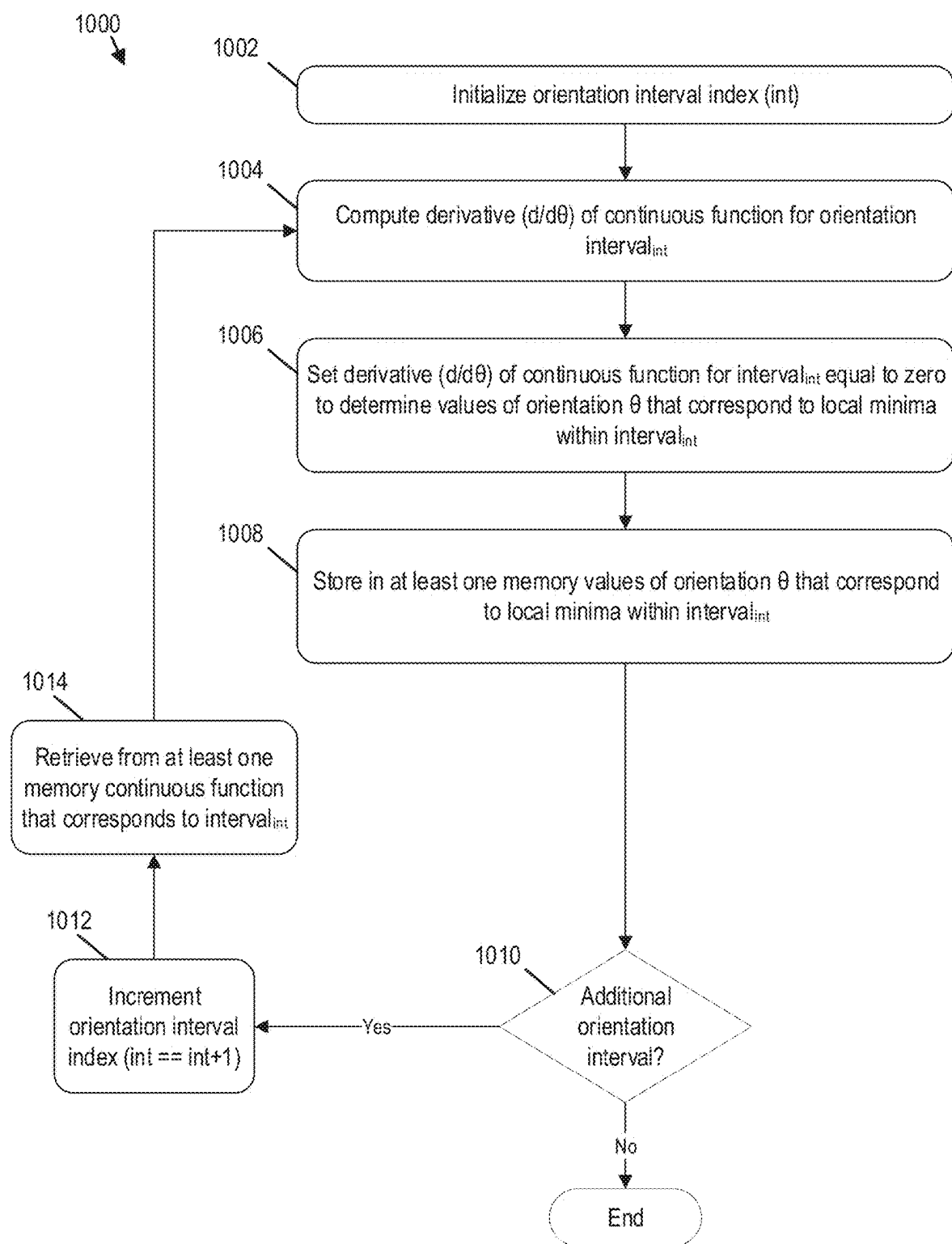


FIG. 10

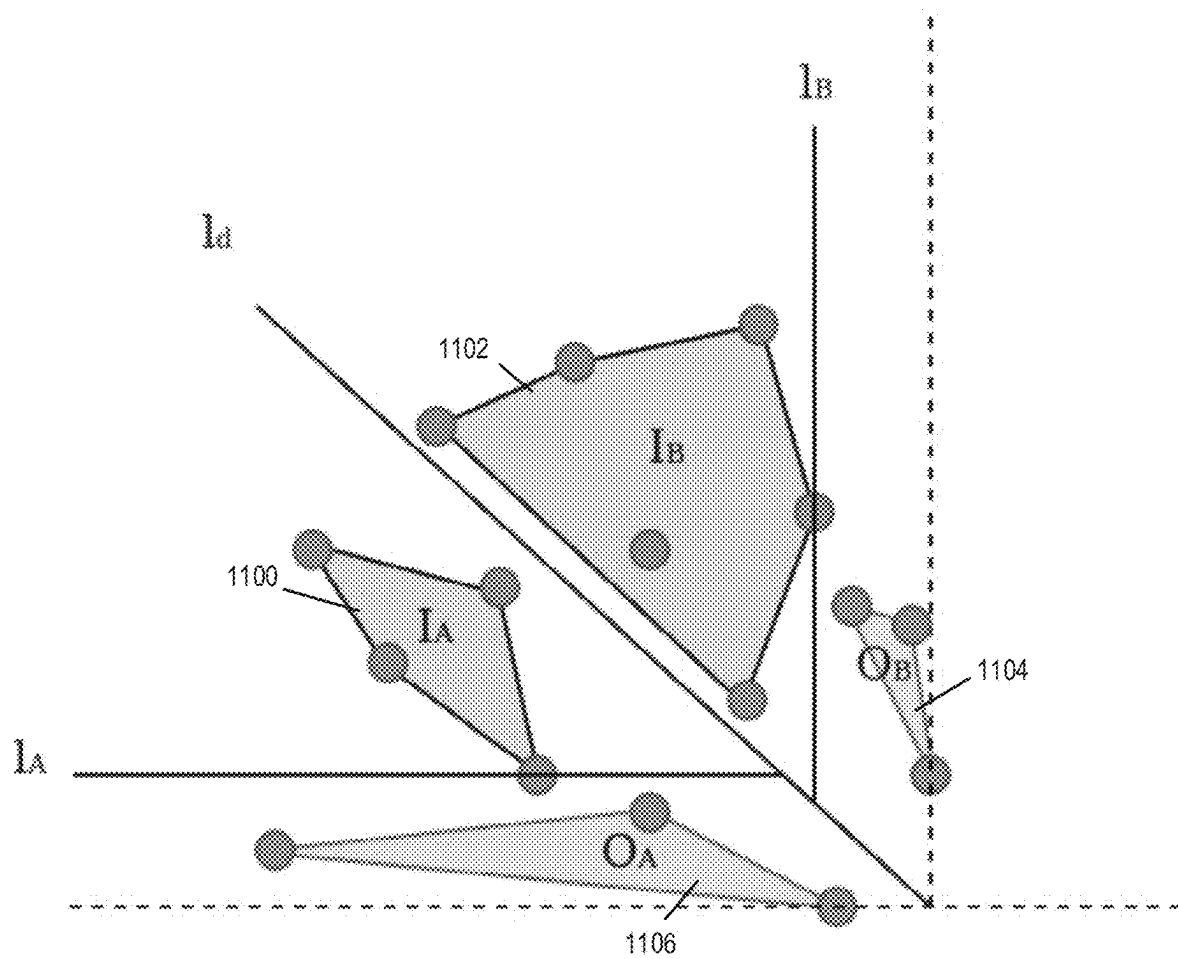


FIG. 11

SYSTEMS AND METHODS FOR EFFICIENT VEHICLE EXTENT ESTIMATION

RELATED APPLICATIONS

[0001] Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are incorporated by reference under 37 CFR 1.57 and made a part of this specification. This application is a continuation of U.S. patent application Ser. No. 17/804,522, filed on May 27, 2022, and entitled SYSTEMS AND METHODS FOR EFFICIENT VEHICLE EXTENT ESTIMATION, which claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 63/229,370, filed on Aug. 4, 2021, and entitled SYSTEMS AND METHODS FOR EFFICIENT VEHICLE EXTENT ESTIMATION, each of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Current vehicle extent estimation methods commonly employ remote sensing measurements, such as light detection and ranging (LiDAR) measurements, to estimate bounding boxes enclosing vehicle outer surfaces. Bounding box orientation estimation is typically accomplished via non-linear optimization methods and a grid search approach. These approaches have their drawbacks, though. A popular approach to shape fitting bounding boxes to measurement points involves minimizing the average distance of the points from sides of the bounding box, for each of a set of possible box orientations. The enclosing bounding box corresponding to the optimal orientation constitutes the fitted shape to the point cluster. This approach is computationally complex and slow, however, involving sorting all measurement points and re-computing their distances for every orientation.

BRIEF DESCRIPTION OF THE FIGURES

[0003] FIG. 1 is an example environment in which a vehicle including one or more components of an autonomous system can be implemented;

[0004] FIG. 2 is a diagram of one or more systems of a vehicle including an autonomous system;

[0005] FIG. 3 is a diagram of components of one or more devices and/or one or more systems of FIGS. 1 and 2;

[0006] FIG. 4 is a diagram of certain components of an autonomous system;

[0007] FIG. 5 is a diagram of an implementation of a process for efficient vehicle extent estimation;

[0008] FIGS. 6A-6G are more detailed diagrams of an implementation of a process for efficient vehicle extent estimation;

[0009] FIGS. 7-10 are flowcharts of processes for efficient vehicle extent estimation; and

[0010] FIG. 11 conceptually illustrates exemplary rejection of outlying data points.

DETAILED DESCRIPTION

[0011] In the following description numerous specific details are set forth in order to provide a thorough understanding of the present disclosure for the purposes of explanation. It will be apparent, however, that the embodiments described by the present disclosure can be practiced without these specific details. In some instances, well-known struc-

tures and devices are illustrated in block diagram form in order to avoid unnecessarily obscuring aspects of the present disclosure.

[0012] Specific arrangements or orderings of schematic elements, such as those representing systems, devices, modules, instruction blocks, data elements, and/or the like are illustrated in the drawings for ease of description. However, it will be understood by those skilled in the art that the specific ordering or arrangement of the schematic elements in the drawings is not meant to imply that a particular order or sequence of processing, or separation of processes, is required unless explicitly described as such. Further, the inclusion of a schematic element in a drawing is not meant to imply that such element is required in all embodiments or that the features represented by such element may not be included in or combined with other elements in some embodiments unless explicitly described as such.

[0013] Further, where connecting elements such as solid or dashed lines or arrows are used in the drawings to illustrate a connection, relationship, or association between or among two or more other schematic elements, the absence of any such connecting elements is not meant to imply that no connection, relationship, or association can exist. In other words, some connections, relationships, or associations between elements are not illustrated in the drawings so as not to obscure the disclosure. In addition, for ease of illustration, a single connecting element can be used to represent multiple connections, relationships, or associations between elements. For example, where a connecting element represents communication of signals, data, or instructions (e.g., “software instructions”), it should be understood by those skilled in the art that such element can represent one or multiple signal paths (e.g., a bus), as may be needed, to affect the communication.

[0014] Although the terms first, second, third, and/or the like are used to describe various elements, these elements should not be limited by these terms. The terms first, second, third, and/or the like are used only to distinguish one element from another. For example, a first contact could be termed a second contact and, similarly, a second contact could be termed a first contact without departing from the scope of the described embodiments. The first contact and the second contact are both contacts, but they are not the same contact.

[0015] The terminology used in the description of the various described embodiments herein is included for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a,” “an” and “the” are intended to include the plural forms as well and can be used interchangeably with “one or more” or “at least one,” unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this description specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0016] As used herein, the terms “communication” and “communicate” refer to at least one of the reception, receipt,

transmission, transfer, provision, and/or the like of information (or information represented by, for example, data, signals, messages, instructions, commands, and/or the like). For one unit (e.g., a device, a system, a component of a device or system, combinations thereof, and/or the like) to be in communication with another unit means that the one unit is able to directly or indirectly receive information from and/or send (e.g., transmit) information to the other unit. This may refer to a direct or indirect connection that is wired and/or wireless in nature. Additionally, two units may be in communication with each other even though the information transmitted may be modified, processed, relayed, and/or routed between the first and second unit. For example, a first unit may be in communication with a second unit even though the first unit passively receives information and does not actively transmit information to the second unit. As another example, a first unit may be in communication with a second unit if at least one intermediary unit (e.g., a third unit located between the first unit and the second unit) processes information received from the first unit and transmits the processed information to the second unit. In some embodiments, a message may refer to a network packet (e.g., a data packet and/or the like) that includes data.

[0017] As used herein, the term “if” is, optionally, construed to mean “when”, “upon”, “in response to determining,” “in response to detecting,” and/or the like, depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining,” “in response to determining,” “upon detecting [the stated condition or event],” “in response to detecting [the stated condition or event],” and/or the like, depending on the context. Also, as used herein, the terms “has”, “have”, “having”, or the like are intended to be open-ended terms. Further, the phrase “based on” is intended to mean “based at least partially on” unless explicitly stated otherwise.

[0018] Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the various described embodiments. However, it will be apparent to one of ordinary skill in the art that the various described embodiments can be practiced without these specific details. In other instances, well-known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.

[0019] In some aspects and/or embodiments, systems, methods, and computer program products described herein include and/or implement efficient vehicle extent estimation including bounding box generation. Convex hulls of measurement points are employed to identify bounding box orientations at which one or more points switch from being nearest to one side of the bounding box, to being nearest another side. Between these orientations, a candidate orientation is determined which yields a local minimum distance between the points and the sides of the bounding box. The orientation corresponding to the global minimum is selected from among these local minimum orientations, yielding the bounding box orientation that best fits the measurement points.

[0020] By virtue of the implementation of systems, methods, and computer program products described herein, techniques of embodiments of the disclosure accomplish more

efficient vehicle extent estimation than is possible with prior systems and methods. Processes of embodiments of the disclosure allow for determination of bounding boxes from convex hulls rather than re-sorting and re-calculating point positions for each candidate bounding box orientation, and optimize candidate orientations in continuum, thus using fewer computational resources. Such processes also reduce loss of accuracy from downsampling, by leveraging all measurement points, thus yielding greater heading accuracy.

[0021] Referring now to FIG. 1, illustrated is example environment 100 in which vehicles that include autonomous systems, as well as vehicles that do not, are operated. As illustrated, environment 100 includes vehicles 102a-102n, objects 104a-104n, routes 106a-106n, area 108, vehicle-to-infrastructure (V2I) device 110, network 112, remote autonomous vehicle (AV) system 114, fleet management system 116, and V2I system 118. Vehicles 102a-102n, vehicle-to-infrastructure (V2I) device 110, network 112, autonomous vehicle (AV) system 114, fleet management system 116, and V2I system 118 interconnect (e.g., establish a connection to communicate and/or the like) via wired connections, wireless connections, or a combination of wired or wireless connections. In some embodiments, objects 104a-104n interconnect with at least one of vehicles 102a-102n, vehicle-to-infrastructure (V2I) device 110, network 112, autonomous vehicle (AV) system 114, fleet management system 116, and V2I system 118 via wired connections, wireless connections, or a combination of wired or wireless connections.

[0022] Vehicles 102a-102n (referred to individually as vehicle 102 and collectively as vehicles 102) include at least one device configured to transport goods and/or people. In some embodiments, vehicles 102 are configured to be in communication with V2I device 110, remote AV system 114, fleet management system 116, and/or V2I system 118 via network 112. In some embodiments, vehicles 102 include cars, buses, trucks, trains, and/or the like. In some embodiments, vehicles 102 are the same as, or similar to, vehicles 200, described herein (see FIG. 2). In some embodiments, a vehicle 200 of a set of vehicles 200 is associated with an autonomous fleet manager. In some embodiments, vehicles 102 travel along respective routes 106a-106n (referred to individually as route 106 and collectively as routes 106), as described herein. In some embodiments, one or more vehicles 102 include an autonomous system (e.g., an autonomous system that is the same as or similar to autonomous system 202).

[0023] Objects 104a-104n (referred to individually as object 104 and collectively as objects 104) include, for example, at least one vehicle, at least one pedestrian, at least one cyclist, at least one structure (e.g., a building, a sign, a fire hydrant, etc.), a portion of a roadway, a roadside object, and/or the like. Each object 104 is stationary (e.g., located at a fixed location for a period of time) or mobile (e.g., having a velocity and associated with at least one trajectory). In some embodiments, objects 104 are associated with corresponding locations in area 108.

[0024] Routes 106a-106n (referred to individually as route 106 and collectively as routes 106) are each associated with (e.g., prescribe) a sequence of actions (also known as a trajectory) connecting states along which an AV can navigate. Each route 106 starts at an initial state (e.g., a state that corresponds to a first spatiotemporal location, velocity, and/or the like) and a final goal state (e.g., a state that

corresponds to a second spatiotemporal location that is different from the first spatiotemporal location) or goal region (e.g., a subspace of acceptable states (e.g., terminal states)). In some embodiments, the first state includes a location at which an individual or individuals are to be picked-up by the AV and the second state or region includes a location or locations at which the individual or individuals picked-up by the AV are to be dropped-off. In some embodiments, routes **106** include a plurality of acceptable state sequences (e.g., a plurality of spatiotemporal location sequences), the plurality of state sequences associated with (e.g., defining) a plurality of trajectories. In an example, routes **106** include only high level actions or imprecise state locations, such as a series of connected roads dictating turning directions at roadway intersections. Additionally, or alternatively, routes **106** may include more precise actions or states such as, for example, specific target lanes or precise locations within the lane areas and targeted speed at those positions. In an example, routes **106** include a plurality of precise state sequences along the at least one high level action sequence with a limited lookahead horizon to reach intermediate goals, where the combination of successive iterations of limited horizon state sequences cumulatively correspond to a plurality of trajectories that collectively form the high level route to terminate at the final goal state or region.

[0025] Area **108** includes a physical area (e.g., a geographic region) within which vehicles **102** can navigate. In an example, area **108** includes at least one state (e.g., a country, a province, an individual state of a plurality of states included in a country, etc.), at least one portion of a state, at least one city, at least one portion of a city, etc. In some embodiments, area **108** includes at least one named thoroughfare (referred to herein as a “road”) such as a highway, an interstate highway, a parkway, a city street, etc. Additionally, or alternatively, in some examples area **108** includes at least one unnamed road such as a driveway, a section of a parking lot, a section of a vacant and/or undeveloped lot, a dirt path, etc. In some embodiments, a road includes at least one lane (e.g., a portion of the road that can be traversed by vehicles **102**). In an example, a road includes at least one lane associated with (e.g., identified based on) at least one lane marking.

[0026] Vehicle-to-Infrastructure (V2I) device **110** (sometimes referred to as a Vehicle-to-Infrastructure (V2X) device) includes at least one device configured to be in communication with vehicles **102** and/or V2I infrastructure system **118**. In some embodiments, V2I device **110** is configured to be in communication with vehicles **102**, remote AV system **114**, fleet management system **116**, and/or V2I system **118** via network **112**. In some embodiments, V2I device **110** includes a radio frequency identification (RFID) device, signage, cameras (e.g., two-dimensional (2D) and/or three-dimensional (3D) cameras), lane markers, streetlights, parking meters, etc. In some embodiments, V2I device **110** is configured to communicate directly with vehicles **102**. Additionally, or alternatively, in some embodiments V2I device **110** is configured to communicate with vehicles **102**, remote AV system **114**, and/or fleet management system **116** via V2I system **118**. In some embodiments, V2I device **110** is configured to communicate with V2I system **118** via network **112**.

[0027] Network **112** includes one or more wired and/or wireless networks. In an example, network **112** includes a

cellular network (e.g., a long term evolution (LTE) network, a third generation (3G) network, a fourth generation (4G) network, a fifth generation (5G) network, a code division multiple access (CDMA) network, etc.), a public land mobile network (PLMN), a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), a telephone network (e.g., the public switched telephone network (PSTN)), a private network, an ad hoc network, an intranet, the Internet, a fiber optic-based network, a cloud computing network, etc., a combination of some or all of these networks, and/or the like.

[0028] Remote AV system **114** includes at least one device configured to be in communication with vehicles **102**, V2I device **110**, network **112**, remote AV system **114**, fleet management system **116**, and/or V2I system **118** via network **112**. In an example, remote AV system **114** includes a server, a group of servers, and/or other like devices. In some embodiments, remote AV system **114** is co-located with the fleet management system **116**. In some embodiments, remote AV system **114** is involved in the installation of some or all of the components of a vehicle, including an autonomous system, an autonomous vehicle compute, software implemented by an autonomous vehicle compute, and/or the like. In some embodiments, remote AV system **114** maintains (e.g., updates and/or replaces) such components and/or software during the lifetime of the vehicle.

[0029] Fleet management system **116** includes at least one device configured to be in communication with vehicles **102**, V2I device **110**, remote AV system **114**, and/or V2I infrastructure system **118**. In an example, fleet management system **116** includes a server, a group of servers, and/or other like devices. In some embodiments, fleet management system **116** is associated with a ridesharing company (e.g., an organization that controls operation of multiple vehicles (e.g., vehicles that include autonomous systems and/or vehicles that do not include autonomous systems) and/or the like).

[0030] In some embodiments, V2I system **118** includes at least one device configured to be in communication with vehicles **102**, V2I device **110**, remote AV system **114**, and/or fleet management system **116** via network **112**. In some examples, V2I system **118** is configured to be in communication with V2I device **110** via a connection different from network **112**. In some embodiments, V2I system **118** includes a server, a group of servers, and/or other like devices. In some embodiments, V2I system **118** is associated with a municipality or a private institution (e.g., a private institution that maintains V2I device **110** and/or the like).

[0031] The number and arrangement of elements illustrated in FIG. 1 are provided as an example. There can be additional elements, fewer elements, different elements, and/or differently arranged elements, than those illustrated in FIG. 1. Additionally, or alternatively, at least one element of environment **100** can perform one or more functions described as being performed by at least one different element of FIG. 1. Additionally, or alternatively, at least one set of elements of environment **100** can perform one or more functions described as being performed by at least one different set of elements of environment **100**.

[0032] Referring now to FIG. 2, vehicle **200** includes autonomous system **202**, powertrain control system **204**, steering control system **206**, and brake system **208**. In some embodiments, vehicle **200** is the same as or similar to vehicle **102** (see FIG. 1). In some embodiments, vehicle **102**

have autonomous capability (e.g., implement at least one function, feature, device, and/or the like that enable vehicle **200** to be partially or fully operated without human intervention including, without limitation, fully autonomous vehicles (e.g., vehicles that forego reliance on human intervention), highly autonomous vehicles (e.g., vehicles that forego reliance on human intervention in certain situations), and/or the like). For a detailed description of fully autonomous vehicles and highly autonomous vehicles, reference may be made to SAE International's standard J3016: Taxonomy and Definitions for Terms Related to On-Road Motor Vehicle Automated Driving Systems, which is incorporated by reference in its entirety. In some embodiments, vehicle **200** is associated with an autonomous fleet manager and/or a ridesharing company.

[0033] Autonomous system **202** includes a sensor suite that includes one or more devices such as cameras **202a**, LiDAR sensors **202b**, radar sensors **202c**, and microphones **202d**. In some embodiments, autonomous system **202** can include more or fewer devices and/or different devices (e.g., ultrasonic sensors, inertial sensors, GPS receivers (discussed below), odometry sensors that generate data associated with an indication of a distance that vehicle **200** has traveled, and/or the like). In some embodiments, autonomous system **202** uses the one or more devices included in autonomous system **202** to generate data associated with environment **100**, described herein. The data generated by the one or more devices of autonomous system **202** can be used by one or more systems described herein to observe the environment (e.g., environment **100**) in which vehicle **200** is located. In some embodiments, autonomous system **202** includes communication device **202e**, autonomous vehicle compute **202f**, and drive-by-wire (DBW) system **202h**.

[0034] Cameras **202a** include at least one device configured to be in communication with communication device **202e**, autonomous vehicle compute **202f**, and/or safety controller **202g** via a bus (e.g., a bus that is the same as or similar to bus **302** of FIG. 3). Cameras **202a** include at least one camera (e.g., a digital camera using a light sensor such as a charge-coupled device (CCD), a thermal camera, an infrared (IR) camera, an event camera, and/or the like) to capture images including physical objects (e.g., cars, buses, curbs, people, and/or the like). In some embodiments, camera **202a** generates camera data as output. In some examples, camera **202a** generates camera data that includes image data associated with an image. In this example, the image data may specify at least one parameter (e.g., image characteristics such as exposure, brightness, etc., an image time-stamp, and/or the like) corresponding to the image. In such an example, the image may be in a format (e.g., RAW, JPEG, PNG, and/or the like). In some embodiments, camera **202a** includes a plurality of independent cameras configured on (e.g., positioned on) a vehicle to capture images for the purpose of stereopsis (stereo vision). In some examples, camera **202a** includes a plurality of cameras that generate image data and transmit the image data to autonomous vehicle compute **202f** and/or a fleet management system (e.g., a fleet management system that is the same as or similar to fleet management system **116** of FIG. 1). In such an example, autonomous vehicle compute **202f** determines depth to one or more objects in a field of view of at least two cameras of the plurality of cameras based on the image data from the at least two cameras. In some embodiments, cameras **202a** is configured to capture images of objects

within a distance from cameras **202a** (e.g., up to 100 meters, up to a kilometer, and/or the like). Accordingly, cameras **202a** include features such as sensors and lenses that are optimized for perceiving objects that are at one or more distances from cameras **202a**.

[0035] In an embodiment, camera **202a** includes at least one camera configured to capture one or more images associated with one or more traffic lights, street signs and/or other physical objects that provide visual navigation information. In some embodiments, camera **202a** generates traffic light data associated with one or more images. In some examples, camera **202a** generates TLD data associated with one or more images that include a format (e.g., RAW, JPEG, PNG, and/or the like). In some embodiments, camera **202a** that generates TLD data differs from other systems described herein incorporating cameras in that camera **202a** can include one or more cameras with a wide field of view (e.g., a wide-angle lens, a fish-eye lens, a lens having a viewing angle of approximately 120 degrees or more, and/or the like) to generate images about as many physical objects as possible.

[0036] LiDAR sensors **202b** include at least one device configured to be in communication with communication device **202e**, autonomous vehicle compute **202f**, and/or safety controller **202g** via a bus (e.g., a bus that is the same as or similar to bus **302** of FIG. 3). LiDAR sensors **202b** include a system configured to transmit light from a light emitter (e.g., a laser transmitter). Light emitted by LiDAR sensors **202b** include light (e.g., infrared light and/or the like) that is outside of the visible spectrum. In some embodiments, during operation, light emitted by LiDAR sensors **202b** encounters a physical object (e.g., a vehicle) and is reflected back to LiDAR sensors **202b**. In some embodiments, the light emitted by LiDAR sensors **202b** does not penetrate the physical objects that the light encounters. LiDAR sensors **202b** also include at least one light detector which detects the light that was emitted from the light emitter after the light encounters a physical object. In some embodiments, at least one data processing system associated with LiDAR sensors **202b** generates an image (e.g., a point cloud, a combined point cloud, and/or the like) representing the objects included in a field of view of LiDAR sensors **202b**. In some examples, the at least one data processing system associated with LiDAR sensor **202b** generates an image that represents the boundaries of a physical object, the surfaces (e.g., the topology of the surfaces) of the physical object, and/or the like. In such an example, the image is used to determine the boundaries of physical objects in the field of view of LiDAR sensors **202b**.

[0037] Radio Detection and Ranging (radar) sensors **202c** include at least one device configured to be in communication with communication device **202e**, autonomous vehicle compute **202f**, and/or safety controller **202g** via a bus (e.g., a bus that is the same as or similar to bus **302** of FIG. 3). Radar sensors **202c** include a system configured to transmit radio waves (either pulsed or continuously). The radio waves transmitted by radar sensors **202c** include radio waves that are within a predetermined spectrum. In some embodiments, during operation, radio waves transmitted by radar sensors **202c** encounter a physical object and are reflected back to radar sensors **202c**. In some embodiments, the radio waves transmitted by radar sensors **202c** are not reflected by some objects. In some embodiments, at least one data processing system associated with radar sensors

202c generates signals representing the objects included in a field of view of radar sensors **202c**. For example, the at least one data processing system associated with radar sensor **202c** generates an image that represents the boundaries of a physical object, the surfaces (e.g., the topology of the surfaces) of the physical object, and/or the like. In some examples, the image is used to determine the boundaries of physical objects in the field of view of radar sensors **202c**.

[0038] Microphones **202d** includes at least one device configured to be in communication with communication device **202e**, autonomous vehicle compute **202f**, and/or safety controller **202g** via a bus (e.g., a bus that is the same as or similar to bus **302** of FIG. 3). Microphones **202d** include one or more microphones (e.g., array microphones, external microphones, and/or the like) that capture audio signals and generate data associated with (e.g., representing) the audio signals. In some examples, microphones **202d** include transducer devices and/or like devices. In some embodiments, one or more systems described herein can receive the data generated by microphones **202d** and determine a position of an object relative to vehicle **200** (e.g., a distance and/or the like) based on the audio signals associated with the data.

[0039] Communication device **202e** include at least one device configured to be in communication with cameras **202a**, LiDAR sensors **202b**, radar sensors **202c**, microphones **202d**, autonomous vehicle compute **202f**, safety controller **202g**, and/or DBW system **202h**. For example, communication device **202e** may include a device that is the same as or similar to communication interface **314** of FIG. 3. In some embodiments, communication device **202e** includes a vehicle-to-vehicle (V2V) communication device (e.g., a device that enables wireless communication of data between vehicles).

[0040] Autonomous vehicle compute **202f** includes at least one device configured to be in communication with cameras **202a**, LiDAR sensors **202b**, radar sensors **202c**, microphones **202d**, communication device **202e**, safety controller **202g**, and/or DBW system **202h**. In some examples, autonomous vehicle compute **202f** includes a device such as a client device, a mobile device (e.g., a cellular telephone, a tablet, and/or the like) a server (e.g., a computing device including one or more central processing units, graphical processing units, and/or the like), and/or the like. In some embodiments, autonomous vehicle compute **202f** is the same as or similar to autonomous vehicle compute **400**, described herein. Additionally, or alternatively, in some embodiments autonomous vehicle compute **202f** is configured to be in communication with an autonomous vehicle system (e.g., an autonomous vehicle system that is the same as or similar to remote AV system **114** of FIG. 1), a fleet management system (e.g., a fleet management system that is the same as or similar to fleet management system **116** of FIG. 1), a V2I device (e.g., a V2I device that is the same as or similar to V2I device **110** of FIG. 1), and/or a V2I system (e.g., a V2I system that is the same as or similar to V2I system **118** of FIG. 1).

[0041] Safety controller **202g** includes at least one device configured to be in communication with cameras **202a**, LiDAR sensors **202b**, radar sensors **202c**, microphones **202d**, communication device **202e**, autonomous vehicle computer **202f**, and/or DBW system **202h**. In some examples, safety controller **202g** includes one or more controllers (electrical controllers, electromechanical con-

trollers, and/or the like) that are configured to generate and/or transmit control signals to operate one or more devices of vehicle **200** (e.g., powertrain control system **204**, steering control system **206**, brake system **208**, and/or the like). In some embodiments, safety controller **202g** is configured to generate control signals that take precedence over (e.g., overrides) control signals generated and/or transmitted by autonomous vehicle compute **202f**.

[0042] DBW system **202h** includes at least one device configured to be in communication with communication device **202e** and/or autonomous vehicle compute **202f**. In some examples, DBW system **202h** includes one or more controllers (e.g., electrical controllers, electromechanical controllers, and/or the like) that are configured to generate and/or transmit control signals to operate one or more devices of vehicle **200** (e.g., powertrain control system **204**, steering control system **206**, brake system **208**, and/or the like). Additionally, or alternatively, the one or more controllers of DBW system **202h** are configured to generate and/or transmit control signals to operate at least one different device (e.g., a turn signal, headlights, door locks, windshield wipers, and/or the like) of vehicle **200**.

[0043] Powertrain control system **204** includes at least one device configured to be in communication with DBW system **202h**. In some examples, powertrain control system **204** includes at least one controller, actuator, and/or the like. In some embodiments, powertrain control system **204** receives control signals from DBW system **202h** and powertrain control system **204** causes vehicle **200** to start moving forward, stop moving forward, start moving backward, stop moving backward, accelerate in a direction, decelerate in a direction, perform a left turn, perform a right turn, and/or the like. In an example, powertrain control system **204** causes the energy (e.g., fuel, electricity, and/or the like) provided to a motor of the vehicle to increase, remain the same, or decrease, thereby causing at least one wheel of vehicle **200** to rotate or not rotate.

[0044] Steering control system **206** includes at least one device configured to rotate one or more wheels of vehicle **200**. In some examples, steering control system **206** includes at least one controller, actuator, and/or the like. In some embodiments, steering control system **206** causes the front two wheels and/or the rear two wheels of vehicle **200** to rotate to the left or right to cause vehicle **200** to turn to the left or right.

[0045] Brake system **208** includes at least one device configured to actuate one or more brakes to cause vehicle **200** to reduce speed and/or remain stationary. In some examples, brake system **208** includes at least one controller and/or actuator that is configured to cause one or more calipers associated with one or more wheels of vehicle **200** to close on a corresponding rotor of vehicle **200**. Additionally, or alternatively, in some examples brake system **208** includes an automatic emergency braking (AEB) system, a regenerative braking system, and/or the like.

[0046] In some embodiments, vehicle **200** includes at least one platform sensor (not explicitly illustrated) that measures or infers properties of a state or a condition of vehicle **200**. In some examples, vehicle **200** includes platform sensors such as a global positioning system (GPS) receiver, an inertial measurement unit (IMU), a wheel speed sensor, a wheel brake pressure sensor, a wheel torque sensor, an engine torque sensor, a steering angle sensor, and/or the like.

[0047] Referring now to FIG. 3, illustrated is a schematic diagram of a device 300. As illustrated, device 300 includes processor 304, memory 306, storage component 308, input interface 310, output interface 312, communication interface 314, and bus 302. In some embodiments, device 300 corresponds to at least one device of vehicles 102 (e.g., at least one device of a system of vehicles 102), at least one device of vehicle 200 (e.g., at least one device of a system of vehicle 200), and/or one or more devices of network 112 (e.g., one or more devices of a system of network 112). In some embodiments, one or more devices of vehicles 102 (e.g., one or more devices of a system of vehicles 102), one or more devices of vehicle 200 (e.g., one or more devices of a system of vehicle 200), and/or one or more devices of network 112 (e.g., one or more devices of a system of network 112) include at least one device 300 and/or at least one component of device 300. As shown in FIG. 3, device 300 includes bus 302, processor 304, memory 306, storage component 308, input interface 310, output interface 312, and communication interface 314.

[0048] Bus 302 includes a component that permits communication among the components of device 300. In some cases, processor 304 includes a processor (e.g., a central processing unit (CPU), a graphics processing unit (GPU), an accelerated processing unit (APU), and/or the like), a microphone, a digital signal processor (DSP), and/or any processing component (e.g., a field-programmable gate array (FPGA), an application specific integrated circuit (ASIC), and/or the like) that can be programmed to perform at least one function. Memory 306 includes random access memory (RAM), read-only memory (ROM), and/or another type of dynamic and/or static storage device (e.g., flash memory, magnetic memory, optical memory, and/or the like) that stores data and/or instructions for use by processor 304.

[0049] Storage component 308 stores data and/or software related to the operation and use of device 300. In some examples, storage component 308 includes a hard disk (e.g., a magnetic disk, an optical disk, a magneto-optic disk, a solid state disk, and/or the like), a compact disc (CD), a digital versatile disc (DVD), a floppy disk, a cartridge, a magnetic tape, a CD-ROM, RAM, PROM, EPROM, FLASH-EPROM, NV-RAM, and/or another type of computer readable medium, along with a corresponding drive.

[0050] Input interface 310 includes a component that permits device 300 to receive information, such as via user input (e.g., a touchscreen display, a keyboard, a keypad, a mouse, a button, a switch, a microphone, a camera, and/or the like). Additionally or alternatively, in some embodiments input interface 310 includes a sensor that senses information (e.g., a global positioning system (GPS) receiver, an accelerometer, a gyroscope, an actuator, and/or the like). Output interface 312 includes a component that provides output information from device 300 (e.g., a display, a speaker, one or more light-emitting diodes (LEDs), and/or the like).

[0051] In some embodiments, communication interface 314 includes a transceiver-like component (e.g., a transceiver, a separate receiver and transmitter, and/or the like) that permits device 300 to communicate with other devices via a wired connection, a wireless connection, or a combination of wired and wireless connections. In some examples, communication interface 314 permits device 300 to receive information from another device and/or provide information to another device. In some examples, communication interface 314 includes an Ethernet interface, an optical interface,

a coaxial interface, an infrared interface, a radio frequency (RF) interface, a universal serial bus (USB) interface, a Wi-Fi® interface, a cellular network interface, and/or the like.

[0052] In some embodiments, device 300 performs one or more processes described herein.

[0053] Device 300 performs these processes based on processor 304 executing software instructions stored by a computer-readable medium, such as memory 305 and/or storage component 308. A computer-readable medium (e.g., a non-transitory computer readable medium) is defined herein as a non-transitory memory device. A non-transitory memory device includes memory space located inside a single physical storage device or memory space spread across multiple physical storage devices.

[0054] In some embodiments, software instructions are read into memory 306 and/or storage component 308 from another computer-readable medium or from another device via communication interface 314. When executed, software instructions stored in memory 306 and/or storage component 308 cause processor 304 to perform one or more processes described herein. Additionally or alternatively, hardwired circuitry is used in place of or in combination with software instructions to perform one or more processes described herein. Thus, embodiments described herein are not limited to any specific combination of hardware circuitry and software unless explicitly stated otherwise.

[0055] Memory 306 and/or storage component 308 includes data storage or at least one data structure (e.g., a database and/or the like). Device 300 is capable of receiving information from, storing information in, communicating information to, or searching information stored in the data storage or the at least one data structure in memory 306 or storage component 308. In some examples, the information includes network data, input data, output data, or any combination thereof.

[0056] In some embodiments, device 300 is configured to execute software instructions that are either stored in memory 306 and/or in the memory of another device (e.g., another device that is the same as or similar to device 300). As used herein, the term “module” refers to at least one instruction stored in memory 306 and/or in the memory of another device that, when executed by processor 304 and/or by a processor of another device (e.g., another device that is the same as or similar to device 300) cause device 300 (e.g., at least one component of device 300) to perform one or more processes described herein. In some embodiments, a module is implemented in software, firmware, hardware, and/or the like.

[0057] The number and arrangement of components illustrated in FIG. 3 are provided as an example. In some embodiments, device 300 can include additional components, fewer components, different components, or differently arranged components than those illustrated in FIG. 3. Additionally or alternatively, a set of components (e.g., one or more components) of device 300 can perform one or more functions described as being performed by another component or another set of components of device 300.

[0058] Referring now to FIG. 4, illustrated is an example block diagram of an autonomous vehicle compute 400 (sometimes referred to as an “AV stack”). As illustrated, autonomous vehicle compute 400 includes perception system 402 (sometimes referred to as a perception module), planning system 404 (sometimes referred to as a planning

module), localization system **406** (sometimes referred to as a localization module), control system **408** (sometimes referred to as a control module), and database **410**. In some embodiments, perception system **402**, planning system **404**, localization system **406**, control system **408**, and database **410** are included and/or implemented in an autonomous navigation system of a vehicle (e.g., autonomous vehicle compute **202f** of vehicle **200**). Additionally, or alternatively, in some embodiments perception system **402**, planning system **404**, localization system **406**, control system **408**, and database **410** are included in one or more standalone systems (e.g., one or more systems that are the same as or similar to autonomous vehicle compute **400** and/or the like). In some examples, perception system **402**, planning system **404**, localization system **406**, control system **408**, and database **410** are included in one or more standalone systems that are located in a vehicle and/or at least one remote system as described herein. In some embodiments, any and/or all of the systems included in autonomous vehicle compute **400** are implemented in software (e.g., in software instructions stored in memory), computer hardware (e.g., by microprocessors, microcontrollers, application-specific integrated circuits [ASICs], Field Programmable Gate Arrays (FPGAs), and/or the like), or combinations of computer software and computer hardware. It will also be understood that, in some embodiments, autonomous vehicle compute **400** is configured to be in communication with a remote system (e.g., an autonomous vehicle system that is the same as or similar to remote AV system **114**, a fleet management system **116** that is the same as or similar to fleet management system **116**, a V2I system that is the same as or similar to V2I system **118**, and/or the like).

[0059] In some embodiments, perception system **402** receives data associated with at least one physical object (e.g., data that is used by perception system **402** to detect the at least one physical object) in an environment and classifies the at least one physical object. In some examples, perception system **402** receives image data captured by at least one camera (e.g., cameras **202a**), the image associated with (e.g., representing) one or more physical objects within a field of view of the at least one camera. In such an example, perception system **402** classifies at least one physical object based on one or more groupings of physical objects (e.g., bicycles, vehicles, traffic signs, pedestrians, and/or the like). In some embodiments, perception system **402** transmits data associated with the classification of the physical objects to planning system **404** based on perception system **402** classifying the physical objects.

[0060] In some embodiments, planning system **404** receives data associated with a destination and generates data associated with at least one route (e.g., routes **106**) along which a vehicle (e.g., vehicles **102**) can travel along toward a destination. In some embodiments, planning system **404** periodically or continuously receives data from perception system **402** (e.g., data associated with the classification of physical objects, described above) and planning system **404** updates the at least one trajectory or generates at least one different trajectory based on the data generated by perception system **402**. In some embodiments, planning system **404** receives data associated with an updated position of a vehicle (e.g., vehicles **102**) from localization system **406** and planning system **404** updates the at least one trajectory or generates at least one different trajectory based on the data generated by localization system **406**.

[0061] In some embodiments, localization system **406** receives data associated with (e.g., representing) a location of a vehicle (e.g., vehicles **102**) in an area. In some examples, localization system **406** receives LiDAR data associated with at least one point cloud generated by at least one LiDAR sensor (e.g., LiDAR sensors **202b**). In certain examples, localization system **406** receives data associated with at least one point cloud from multiple LiDAR sensors and localization system **406** generates a combined point cloud based on each of the point clouds. In these examples, localization system **406** compares the at least one point cloud or the combined point cloud to two-dimensional (2D) and/or a three-dimensional (3D) map of the area stored in database **410**. Localization system **406** then determines the position of the vehicle in the area based on localization system **406** comparing the at least one point cloud or the combined point cloud to the map. In some embodiments, the map includes a combined point cloud of the area generated prior to navigation of the vehicle. In some embodiments, maps include, without limitation, high-precision maps of the roadway geometric properties, maps describing road network connectivity properties, maps describing roadway physical properties (such as traffic speed, traffic volume, the number of vehicular and cyclist traffic lanes, lane width, lane traffic directions, or lane marker types and locations, or combinations thereof), and maps describing the spatial locations of road features such as crosswalks, traffic signs or other travel signals of various types. In some embodiments, the map is generated in real-time based on the data received by the perception system.

[0062] In another example, localization system **406** receives Global Navigation Satellite System (GNSS) data generated by a global positioning system (GPS) receiver. In some examples, localization system **406** receives GNSS data associated with the location of the vehicle in the area and localization system **406** determines a latitude and longitude of the vehicle in the area. In such an example, localization system **406** determines the position of the vehicle in the area based on the latitude and longitude of the vehicle. In some embodiments, localization system **406** generates data associated with the position of the vehicle. In some examples, localization system **406** generates data associated with the position of the vehicle based on localization system **406** determining the position of the vehicle. In such an example, the data associated with the position of the vehicle includes data associated with one or more semantic properties corresponding to the position of the vehicle.

[0063] In some embodiments, control system **408** receives data associated with at least one trajectory from planning system **404** and control system **408** controls operation of the vehicle. In some examples, control system **408** receives data associated with at least one trajectory from planning system **404** and control system **408** controls operation of the vehicle by generating and transmitting control signals to cause a powertrain control system (e.g., DBW system **202h**, powertrain control system **204**, and/or the like), a steering control system (e.g., steering control system **206**), and/or a brake system (e.g., brake system **208**) to operate. In an example, where a trajectory includes a left turn, control system **408** transmits a control signal to cause steering control system **206** to adjust a steering angle of vehicle **200**, thereby causing vehicle **200** to turn left. Additionally, or alternatively, control system **408** generates and transmits

control signals to cause other devices (e.g., headlights, turn signal, door locks, windshield wipers, and/or the like) of vehicle 200 to change states.

[0064] In some embodiments, perception system 402, planning system 404, localization system 406, and/or control system 408 implement at least one machine learning model (e.g., at least one multilayer perceptron (MLP), at least one convolutional neural network (CNN), at least one recurrent neural network (RNN), at least one autoencoder, at least one transformer, and/or the like). In some examples, perception system 402, planning system 404, localization system 406, and/or control system 408 implement at least one machine learning model alone or in combination with one or more of the above-noted systems. In some examples, perception system 402, planning system 404, localization system 406, and/or control system 408 implement at least one machine learning model as part of a pipeline (e.g., a pipeline for identifying one or more objects located in an environment and/or the like). An example of an implementation of a machine learning model is included below with respect to FIGS. 4B-4D.

[0065] Database 410 stores data that is transmitted to, received from, and/or updated by perception system 402, planning system 404, localization system 406 and/or control system 408. In some examples, database 410 includes a storage component (e.g., a storage component that is the same as or similar to storage component 308 of FIG. 3) that stores data and/or software related to the operation and uses at least one system of autonomous vehicle compute 400. In some embodiments, database 410 stores data associated with 2D and/or 3D maps of at least one area. In some examples, database 410 stores data associated with 2D and/or 3D maps of a portion of a city, multiple portions of multiple cities, multiple cities, a county, a state, a State (e.g., a country), and/or the like). In such an example, a vehicle (e.g., a vehicle that is the same as or similar to vehicles 102 and/or vehicle 200) can drive along one or more drivable regions (e.g., single-lane roads, multi-lane roads, highways, back roads, off road trails, and/or the like) and cause at least one LiDAR sensor (e.g., a LiDAR sensor that is the same as or similar to LiDAR sensors 202b) to generate data associated with an image representing the objects included in a field of view of the at least one LiDAR sensor.

[0066] In some embodiments, database 410 can be implemented across a plurality of devices. In some examples, database 410 is included in a vehicle (e.g., a vehicle that is the same as or similar to vehicles 102 and/or vehicle 200), an autonomous vehicle system (e.g., an autonomous vehicle system that is the same as or similar to remote AV system 114, a fleet management system (e.g., a fleet management system that is the same as or similar to fleet management system 116 of FIG. 1, a V2I system (e.g., a V2I system that is the same as or similar to V2I system 118 of FIG. 1) and/or the like.

[0067] Referring now to FIG. 5, illustrated is a diagram of an implementation 500 of a process for efficient vehicle extent estimation. In some embodiments, this implementation involves perception system 402. As shown in FIG. 5, perception system 402 may receive data associated with a point cluster 502 such as from LiDAR sensors 202b, representative of a nearby vehicle or other object 104n. Perception system 402 generates a corresponding bounding box 504, to surround point cluster 502. Perception system 402 then sorts the received points 506 into two groups, according

to which bounding box side each point is nearest. Perception system 402 then generates the convex hull 508 for each of these groups, and determines the orientation θ intervals 510 for which the two sorted point groups remain constant. For example, perception system 402 sweeps through the different bounding box orientations to determine those orientation ranges for which every point remains closest to the same bounding box side. Those orientations $\theta_1, \theta_2, \theta_3, \dots$, for which a point switches from being closest to one side to being closest to a different side, mark the end of one such interval and the beginning of the next, as shown in the leftmost table of FIG. 5. Perception system 402 then determines locally minimizing orientations 512, or orientations within each orientation interval at which the point cluster is at minimum average distance from their nearest bounding box sides. This produces a local minimum orientation $\theta_{LM1}, \theta_{LM2}, \theta_{LM3}, \dots$, for each orientation interval. The globally minimizing orientation θ_{GM} is then selected 514 from among the local minimum orientations $\theta_{LM1}, \theta_{LM2}, \theta_{LM3}, \dots$, and is simply the local minimum orientation $\theta_{LM1}, \theta_{LM2}, \theta_{LM3}, \dots$ having the lowest average point distance. This orientation θ_{GM} is considered as the orientation of the best-fit bounding box, and a bounding box having this orientation θ_{GM} may accordingly be constructed to surround this point cluster.

[0068] FIGS. 6A-6G are diagrams illustrating the implementation of FIG. 5 in further detail. Referring now to FIG. 6A, an autonomous vehicle 102 may employ one or more LiDAR sensors 202b to detect discrete points 604 corresponding to location values of at least one detected portion of a nearby object such as a vehicle 104. The cluster of points 604 detected from vehicle 104 at a particular time interval may be referred to as a point cloud 602. The point cloud 602 is transmitted as data to be received by perception system 402. Points 604 may convey any desired position information. For example, points 604 may be any two-dimensional coordinates determined by any one or more sensors, and may be measured relative to a reference point or reference axis associated with a vehicle upon which the one or more sensors are placed, e.g., autonomous vehicle 102. Additionally, while the cluster of points 604 corresponding to one or more portion(s) of vehicle 104 is described here as being a cluster of LiDAR measurements, it is noted that the cluster of points 604 may include position information obtained from any source or combination of sources, such as radar measurements, location values determined from one or more images such as visible light or infrared images, ultrasound measurements from an ultrasonic sensor device (not separately shown in the figures), and/or the like.

[0069] With reference to FIG. 6B, perception system 402 generates an initial bounding box surrounding the points 604 of point cloud 602. In particular, system 402 first generates a bounding box 608 with sides L_A and L_B , oriented at an angle θ with respect to a coordinate frame. Angle θ , in various examples, is defined relative to any suitable coordinate frame, such as a coordinate frame that is associated with autonomous vehicle 102, for instance a coordinate frame based on reference 610, which may correspond to a central axis of autonomous vehicle 102. Here, angle θ may be selected arbitrarily from, e.g., $0^\circ \leq \theta \leq 90^\circ$, and L_A and L_B may be positioned to lie tangent to outer points 604. In this manner, an initial bounding box 608 is generated having dimensions and being oriented to surround each point 604.

[0070] It is noted that, while FIG. 6B illustrates generation of a bounding box, e.g., a rectangle, embodiments of the disclosure contemplate generation of any bounding shape that can enclose or substantially enclose any set of points. For example, the bounding enclosure generated may be a square, a rectangle, any polygon, or the like.

[0071] Referring to FIG. 6C, perception system 402 sorts the points 604 into two sets. One set contains those points 604 which lie closer to side L_A than side L_B , and the other set contains those points 604 which lie closer to side L_B than side L_A . For example, for a line L_D that bisects the included angle between L_A and L_B and that forms an angle θ_D with respect to reference 610, perception system 402 sorts points 604 into one set lying to one side of line L_D , and another set lying to the other side of line L_D . In this manner, line L_D may represent a threshold distance for use in sorting points into one or another of a first set or a second set (e.g. corresponding to convex hull I_A or convex hull I_B , described below in connection with FIG. 6D).

[0072] With reference to FIG. 6D, perception system 402 determines the convex hull of each of these sets of points 604. More specifically, system 402 determines the convex hull I_A of the set of points 604 lying closer to side L_A than side L_B , and the convex hull I_B of the set of points 604 lying closer to side L_B than side L_A .

[0073] As shown in FIG. 6E, perception system 402 next rotates the bounding box sides L_A and L_B while maintaining tangency to the convex hulls I_A and I_B , to determine intervals of orientations θ within which the two sets of points 604 remain constant. For example, bounding box sides L_A and L_B are rotated around convex hulls I_A and I_B while maintaining tangency to the convex hulls I_A and I_B , to determine those orientations $\theta_1, \theta_2, \dots$, or transition orientations, at which L_D intersects a point 604. Intervals of orientations θ are thus the values of that lie between successive orientations θ_1, θ_2 , etc. Accordingly, each orientation interval represents a range of θ for which no point 604 switches from being nearest one side L_A, L_B to being nearest the other side L_B, L_A .

[0074] As described above, previous approaches to bounding box generation commonly involved sorting all measurement points and re-computing their distances for every orientation. For M possible orientations and n data points, this approach results in complexity of order $Mn \log(n)$. In contrast, embodiments of the disclosure avoid re-sorting measurement points and re-computing distances for each orientation, instead computing tangent lines to convex hulls (e.g., tangent line $\tan_1$ of point 604a of convex hull I_A forming an angle θ_1 with respect to reference 610, and tangent line $\tan_2$ of point 604b of convex hull I_B forming an angle θ_2 with respect to reference 610) and only switching one or more points between point sets at transitions between orientation intervals. As tangent line computations and convex hull updates may be performed in $\log(n)$ time for each sorted set of measurement points, approaches of embodiments of the disclosure only result in complexity of order $n \log(n)$, reducing computational overhead and processing time.

[0075] With reference to FIG. 6F, perception system 402 generates and stores, such as in memory 306, a table or other set of these transition orientations 620 ($\theta_1, \theta_2, \theta_3, \dots$), corresponding bounding box orientation intervals 622, and the sets of data points for each interval 624. Each set of data points for an interval 624, in some embodiments, is based on

points 604 and includes a first set of data points 626 corresponding to a first convex hull (e.g., convex hull I_A of FIG. 6D) and a second set of data points 628 corresponding to a second convex hull (e.g., convex hull I_B of FIG. 6D). The first set of data points 626, for example, may include for each data point an x and y coordinate, such as $x_{1,1,i}$ and $y_{1,1,i}$ for point index $i=1$ to $n_{1,1}$, where $n_{1,1}$ represents the number of data points in first set of data points 626 for the corresponding bounding box interval 622. The second set of data points 628, for example, may include for each data point an x and y coordinate, such as $x_{1,2,i}$ and $y_{1,2,i}$ for point index $i=1$ to $n_{1,2}$, where $n_{1,2}$ represents the number of data points in second set of data points 628 for the corresponding bounding box interval 622.

[0076] With reference to FIG. 6G, perception system 402 determines locally minimizing bounding box orientations θ for each interval (e.g., the orientation θ , within each orientation interval 632, for which distances from the first set of points 604 to the first side L_A (e.g., distances 640 shown in FIG. 6C), and the distances from the second set of points 604 to the second side L_B (e.g., distances 642 shown in FIG. 6C), are a minimum). In some embodiments of the disclosure, this is accomplished by determining a continuous function $f(\theta)$ for the sum of point 604 distances to their respective sides L_A, L_B . The local minimum distance may then be determined as the minimum of $f(\theta)$ within each orientation interval. Embodiments of the disclosure contemplate use of any metric, definition, or formulation for distance. As one example, the distances between points 604 and their bounding box sides L_A, L_B may be defined as the L1 norm distance d , or Manhattan distance, which can be expressed, as shown in equation (1) below, as a trigonometric sum for a set of n data points:

$$\begin{aligned} \sum d(p_i) &= \sum |ax_i + by_i + c| \\ &= \sum (ax_i + bx_i + c) \\ &= a \sum x_i + b \sum y_i + nc \\ &= aX + bY + nc \\ &= X \cos(\theta) - Y \sin(\theta) + nc = f(\theta), \end{aligned} \quad (1)$$

where θ represents a bounding box orientation, and i represents a point index, $d(p_i)$ represents a distance from point i to its corresponding side (e.g., nearer side from among side L_A and side L_B).

[0077] For a fixed set of points 604, X and Y are measures of the distances between points p_i and corresponding sides L_A and L_B , and thus remain constant. Accordingly, $f(\theta)$ may be optimized in continuum, by setting its derivative to zero, e.g., $f'(\theta)=0$. Thus, perception system 402 may determine each continuous function $f_1(\theta), f_2(\theta), f_3(\theta), \dots, f_{34}(\theta)$ for each interval of θ , from the above. Perception system 402 may then determine the orientation $\theta_{LM1}, \theta_{LM2}, \theta_{LM3}, \dots$, of each local minimum of $f(\theta)$ 636 for each interval, in continuum. The global minimum orientation 638 is then the θ_{LM} value corresponding to the lowest $f(\theta)$ value. In this manner, perception system 402 determines the best fit bounding box by determining the bounding box orientation resulting in the lowest overall distance, according to whatever distance metric is chosen. As $f(\theta)$ is a continuous function, exact

solutions for local and global minima of $f(\theta)$ may be determined, resulting in greater accuracy in determining best-fit bounding boxes.

[0078] L1 distance is employed above as but one specific example, and embodiments of the disclosure contemplate use of any distance metric suitable for estimating a fit between points and a surrounding bounding enclosure. For instance, any other distance measurement, such as Euclidian distance or the like, may be employed in determination of an average distance between points **604** and bounding box sides L_A , L_B .

[0079] In some embodiments, as an alternative or in addition to minimizing $f(\theta)$ as described above, perception system **402** minimizes an average distance $f_{avg}(\theta)$ of points **604** to bounding box sides L_A , L_B , as given by equation (2):

$$f_{avg}(\theta) = \frac{\sum_{L_A} d(p_i)}{N_{L_A}} + \frac{\sum_{L_B} d(p_i)}{N_{L_B}}, \quad (2)$$

[0080] where N_{L_A} represents a total number of points in a first set of points (e.g., set of points **626** in FIG. 6F, which may correspond to convex hull I_A in FIG. 6E), and N_{L_B} represents a total number of points in a second set of points (e.g., set of points **628** in FIG. 6F, which may correspond to convex hull I_B in FIG. 6E). It may be observed that equation (2) may take the same form as equation (1), with differing values of X and Y.

[0081] Referring now to FIG. 7, illustrated is a flowchart of a process **700** for efficient vehicle extent estimation. In some embodiments, one or more of the steps described with respect to process **700** are performed (e.g., completely, partially, and/or the like) by autonomous vehicle **102**, as above. Additionally or alternatively, in some embodiments one or more steps described with respect to process **700** may be performed (e.g., completely, partially, and/or the like) by another device or group of devices separate from or including autonomous vehicle **102**, such as a remote server (e.g., a remote server that is the same as or similar to remote AV system **114** and/or fleet management system **116** of FIG. 1) carrying out some or all of the above calculations.

[0082] With continued reference to FIG. 7, perception system **402** generates at least two sides of a bounding box enclosing a cluster of points corresponding to location values of at least a portion of a detected vehicle (block **702**). In an example, perception system **402** generates sides L_A and L_B of a bounding box enclosing a cluster **602** of points **604**, as shown in FIGS. 6A and 6B.

[0083] With continued reference to FIG. 7, perception system **402** sorts the points of the point cluster into a first set of points located within a distance (e.g., a threshold distance defined by line L_D in FIG. 6C and/or FIG. 6D) from the first side of the bounding box, and a second set of points located within that distance from the second side (block **704**), e.g., a set made up of points (e.g., points corresponding to convex hull I_A in FIG. 6D) located closer to one side (e.g., side L_A in FIG. 6D), and another set of points (e.g., points corresponding to convex hull I_B in FIG. 6D) made up of points closer to the other side (e.g., side L_B in FIG. 6D). Further details regarding an example sorting of points that may be performed at block **704** are provided above in connection with FIG. 6C and/or FIG. 6D.

[0084] With continued reference to FIG. 7, perception system **402** generates a first convex hull based on this first set of points, and a second convex hull based on the second set of points (block **706**), as described above with reference to FIG. 6D. In an example, perception system **402** generates a first convex hull I_A and a second convex hull I_B , as shown in FIG. 6D.

[0085] With continued reference to FIG. 7, perception system **402** determines a number of intervals of bounding box orientations within which the first and second sets of points remain constant based on a portion of the first convex hull and a portion of the second convex hull (block **708**). In an example, with reference to FIG. 6E, perception system **402** determines tangents to the two convex hulls (e.g., $\tan_1$ having tangent line angle θ_1 with respect to reference **610** and $\tan_2$ having tangent line angle θ_2 with respect to reference **610**, as shown in FIG. 6E) for each orientation of the bounding box (e.g., ranging from an orientation of $\theta=0^\circ$ to an orientation of $\theta=90^\circ$ with respect to reference **610**), and uses the tangents to determine intervals between which a point changes from one set to the other (e.g., from being closer to one bounding box side such as side L_A to being closer to the other such as side L_B). The bounding box orientation intervals determined at block **708** are stored, for example, in a memory, such as memory **306** (e.g., as bounding box orientation intervals **622** of table **618** shown in FIG. 6F and/or as bounding box orientation intervals **632** of table **630** shown in FIG. 6G).

[0086] With continued reference to FIG. 7, perception system **402** determines, for the intervals determined at block **708**, a bounding box orientation which minimizes distances from each point of the first set of points to the first side, and distances from each point of the second set of points to the second side (block **710**). With reference to FIGS. 6F-6G, a continuous function can be determined for the sum of point distances within each interval. The local minima of these continuous functions for each interval thus indicates the best-fit orientation of each interval, and the best-fit orientation having the smallest distance is then the global minimum, or global best-fit orientation **638**, or best-fit bounding box orientation.

[0087] With continued reference to FIG. 7, perception system **402** provides data of this best-fit bounding box to control circuitry of an autonomous vehicle such as vehicle **102**, the data being configured to cause the control circuitry to control operation of the autonomous vehicle (block **712**). For example, perception system **402** describes the best-fit bounding box to systems of the vehicle **102** for any desired purpose, such as assistance in navigation, or the like.

[0088] Referring now to FIG. 8, illustrated is a flowchart of a process **800**, for determining intervals of bounding box orientations (e.g., bounding box orientation intervals **622** shown in table **618** of FIG. 6F and/or bounding box orientation intervals **632** shown in table **630** of FIG. 6G). In some embodiments, process **800** is the same as or similar to the process of block **708** described above in connection with FIG. 7. In some embodiments, one or more of the steps described with respect to process **800** are performed (e.g., completely, partially, and/or the like) by autonomous vehicle **102**, as above. Additionally or alternatively, in some embodiments one or more steps described with respect to process **800** may be performed (e.g., completely, partially, and/or the like) by another device or group of devices separate from or including autonomous vehicle **102**, such as

a remote server carrying out some or all of the above calculations. It is noted that process 800 allows for updating of convex hulls without increasing complexity. As described herein, tangents to convex hulls are employed to determine bounding box orientation intervals 632. Those tangents that define endpoints of orientation intervals also intersect points that switch convex hulls, leading to updates thereto. Editing of point sets and determination of corresponding convex hulls may then be determined in any manner, such as according to processes described in Timothy Chan, *Dynamic Planar Convex Hull Operations in Near-Logarithmic Amortized Time*, 48 J. ACM 1 (2001), which is hereby incorporated by reference in its entirety.

[0089] With continued reference to FIG. 8, prior to execution of block 802, perception system 402 generates an initial bounding box and sort points 604 into two sets according to which leg L_A , L_B of the bounding box is closer, as illustrated in FIG. 6C and as described in connection with blocks 702, 704, 706 of FIG. 7 above. Perception system 402 subsequently determines the convex hull for each set of points 604 (as illustrated in FIG. 6D), then initializes a convex hull index c for each convex hull, and point index i (block 802).

[0090] With continued reference to FIG. 8, perception system 402 computes a tangent line of a point p_i of a convex hull c (block 804). In an example, with reference to FIG. 6E, perception system 402 computes a tangent line $\tan_1$ of point 604a forming a tangent line angle θ_1 with respect to reference 610.

[0091] With continued reference to FIG. 8, perception system 402 determines whether the tangent line angle formed by the tangent line computed for point p_i of convex hull c at block 804 lies within a threshold range (block 806). For example, perception system 402 may determine whether tangent line angle θ_1 is near angle θ_D of line L_D to within a threshold degree of nearness, to determine whether tangent line angle θ_1 corresponds to a transition orientation (e.g., a bounding box orientation at which point p_i is expected to cross over line L_D and switch from being closer to line L_A to being closer to line L_B).

[0092] With continued reference to FIG. 8, if perception system 402 determines at block 806 that the tangent line does not lie within a threshold range (“No” at block 806), then perception system 402 discards the angle of that tangent line as not corresponding to a transition orientation (block 808).

[0093] With continued reference to FIG. 8, if perception system 402 determines at block 806 that the tangent line lies within a threshold range (“Yes” at block 806), then perception system 402 stores the angle of that tangent line in a memory, such as memory 306, to correspond to a transition orientation, along with the corresponding re-sorted sets of points 604 (block 810), accounting for this newly-determined transition point.

[0094] With continued reference to FIG. 8, after block 808 or block 810, as the case may be, perception system 402 determines whether convex hull c includes another point for which a tangent line is to be computed (block 812). In an example, perception system 402 keeps track of each point of convex hull c and performs the determination at block 812 by sequentially stepping through each point of convex hull c.

[0095] With continued reference to FIG. 8, if perception system 402 determines that convex hull c includes another point for which a tangent line is to be computed (“Yes” at

block 812), then perception system 402 selects the other point on convex hull c and increments the data point index i (block 814) and passes control to block 804 to repeat the processes described above for blocks 804, 806, 808, and/or 810 to determine another transition orientation, if it exists.

[0096] With continued reference to FIG. 8, if perception system 402 determines that convex hull c does not include another point for which a tangent line is to be computed (“No” at block 812), then perception system 402 determines (block 816) whether another convex hull (e.g., one of the convex hulls generated at block 706 of FIG. 7) has not yet been analyzed to determine tangent line angles and/or transition orientations according to the processes described above for blocks 802, 804, 806, 808, 810, 812, and/or 814.

[0097] With continued reference to FIG. 8, if perception system 402 determines that another convex hull (e.g., one of the convex hulls generated at block 706 of FIG. 7) remains to be analyzed to determine tangent line angles and/or transition orientations according to the processes described above for blocks 802, 804, 806, 808, 810, 812, and/or 814 (“Yes” at block 816), perception system 402 increments the convex hull index c to correspond to the other convex hull to be analyzed, initializes data point index i, and passes control to block 804 to repeat the processes described above for blocks 804, 806, 808, and/or 810 to determine the transition orientations for the other convex hull (block 818).

[0098] With continued reference to FIG. 8, if perception system 402 determines that no other convex hull (e.g., one of the convex hulls generated at block 706 of FIG. 7) remains to be analyzed to determine tangent line angles and/or transition orientations according to the processes described above for blocks 802, 804, 806, 808, 810, 812, and/or 814 (“No” at block 816), e.g., if both convex hulls generated at block 706 of FIG. 7 have been analyzed, perception system 402 generates (block 816) a table of intervals of bounding box orientations based on the transition angles (e.g., table 618 of FIG. 6F) that were stored over multiple iterations of the process of block 810 (block 820).

[0099] Referring now to FIG. 9, illustrated is a flowchart of a process 900, for determining an orientation of a bounding box that constitutes a best fit for a vehicle extent estimation. In some embodiments, the best fit bounding box orientation is determined as the bounding box orientation (e.g., θ of FIG. 6D) that minimizes (i) the distances (e.g., shortest distances) from points of a first convex hull (e.g., points 604 of convex hull I_A of FIG. 6D) to a first side of the bounding box (e.g., side L_A of FIG. 6D) and (ii) the distances (e.g., shortest distances) from points of a second convex hull (e.g., points 604 of convex hull I_B of FIG. 6D) to a second side of the bounding box (e.g., side L_B of FIG. 6D). In some embodiments, process 900 is the same as or similar to the process of block 710 described above in connection with FIG. 7. In some embodiments, one or more of the steps described with respect to process 900 are performed (e.g., completely, partially, and/or the like) by autonomous vehicle 102, as above. Additionally or alternatively, in some embodiments one or more steps described with respect to process 900 may be performed (e.g., completely, partially, and/or the like) by another device or group of devices separate from or including autonomous vehicle 102, such as a remote server (e.g., a remote server that is the same as or similar to remote AV system 114 and/or fleet management system 116 of FIG. 1) carrying out some or all of the above calculations.

[0100] With continued reference to FIG. 9, prior to execution of block 902, perception system 402 determines bounding box orientation intervals (e.g., bounding box orientation intervals 622 shown in table 618 of FIG. 6F and/or bounding box orientation intervals 632 shown in table 630 of FIG. 6G) within which the sets of points sorted into convex hulls remain constant, in the manner described above in connection with blocks 702, 704, 706, and 708 of FIG. 7. For example, with reference to FIG. 6D, perception system 402 determines each interval of bounding box orientations (e.g., θ of FIG. 6D) for which (i) the points 604 of convex hull I_A remain sorted into convex hull I_A because the points 604 remain closer to bounding box side L_A than to bounding box side L_B (e.g., the points 604 remain on a same side of diagonal line L_D that bisects the angle between line L_A and line L_B) throughout the particular interval of bounding box orientations and (ii) the points 604 of convex hull I_B remain sorted into convex hull I_B because the points 604 remain closer to bounding box side L_B than to bounding box side L_A (e.g., they remain on a same side of diagonal line L_D that bisects the angle between line L_A and line L_B) throughout the particular interval of bounding box orientations.

[0101] With continued reference to FIG. 9, perception system 402 generates a continuous function $f(\theta)$ for each bounding box orientation interval (e.g., each bounding box orientation interval represented in the table generated at block 820 of FIG. 8), based on the re-sorted point sets and describing the distance (as defined by any suitable metric) between the points of each set and the bounding box sides (block 902).

[0102] With continued reference to FIG. 9, perception system 402 determines, for each orientation interval, a locally minimizing orientation associated with the bounding box, the local minimizing orientation being an orientation at which the distance function $f(\theta)$ reaches a minimum within that interval (block 904), including at its extremes. As above, this may be accomplished by setting $f'(\theta)=0$ for each interval and solving the resulting equation in known manner. The result is a bounding box orientation corresponding to a local minimum distance of points 604 in each set to their respective bounding box sides.

[0103] With continued reference to FIG. 9, perception system 402 selects, from the local minimum orientations determined at block 904, the orientation corresponding to the smallest such distance (block 906), which is deemed the bounding box that best fits the points 604.

[0104] Referring now to FIG. 10, illustrated is a flowchart of a process 1000 for determining a locally minimizing bounding box orientation that locally minimizes, within a particular bounding box orientation interval: (i) distances from points of a first convex hull to a first side of the bounding box (e.g., distances 640 from points 604 of convex hull I_A to side L_A) and (ii) distances from points of a second convex hull to a second side of the bounding box (e.g., distances 642 from points 604 of hull I_B to side L_B). In some embodiments, process 1000 is the same as or similar to the process of block 904 described above in connection with FIG. 9. In some embodiments, one or more of the steps described with respect to process 1000 are performed (e.g., completely, partially, and/or the like) by autonomous vehicle 102, as above. Additionally or alternatively, in some embodiments one or more steps described with respect to process 1000 may be performed (e.g., completely, partially, and/or the like) by another device or group of devices

separate from or including autonomous vehicle 102, such as a remote server (e.g., a remote server that is the same as or similar to remote AV system 114 and/or fleet management system 116 of FIG. 1) carrying out some or all of the above calculations.

[0105] With continued reference to FIG. 10, prior to execution of block 1002, perception system 402 generates a continuous function $f(\theta)$ for a bounding box interval that is based on sets of points of convex hulls and that describes the distance (as defined by any suitable metric) between the points of each set and a nearby bounding box side (e.g., as described above in connection with block 902 of FIG. 9). An example of such a continuous function is provided above as equation (1).

[0106] With continued reference to FIG. 10, perception system 402 initializes an orientation interval index int (block 1002). In an example, perception system 402 utilizes orientation interval index int as an index for the bounding box orientation intervals 622 represented in table 618.

[0107] With continued reference to FIG. 10, perception system 402 computes a derivative with respect to θ of the continuous function $f(\theta)$ for that orientation interval (block 1004), sets the derivative that was computed at block 1004 equal to zero (0), and solves the equation to determine the value of orientation θ that corresponds to the minimum value of $f(\theta)$ within this interval int (block 1006).

[0108] With continued reference to FIG. 10, perception system 402 stores the value of θ determined at block 1006, which corresponds to the local minimum of $f(\theta)$ within this bounding box orientation interval, in memory, such as memory 306 (block 1008). In an example, perception system 402 stores the local minimum orientation (e.g., θ_{LM1}) as an entry 636 in a table such as table 630 shown in FIG. 6G.

[0109] With continued reference to FIG. 10, perception system 402 determines whether any additional orientation interval remains to be processed to determine its local minimum of $f(\theta)$ (block 1010). In an example, perception system 402 makes the determination at block 101 by comparing the present value of orientation interval index int to a threshold value (e.g., corresponding to the total number of intervals having entries stored in table 630).

[0110] With continued reference to FIG. 10, if perception system 402 determines that an additional orientation interval remains to be processed to determine its local minimum of $f(\theta)$ ("Yes" at block 1010), perception system 402 increments the orientation interval index int (block 1012).

[0111] With continued reference to FIG. 10, perception system 402 retrieves (block 1014) the continuous function $f(\theta)$ for the interval corresponding to the value of int that was incremented at block 1012, and passes control to block 1004 to repeat the processes described above for blocks 1004, 1006, 1008, and/or 1010 to determine the orientation value θ corresponding to the local minimum value of $f(\theta)$ for this interval.

[0112] With continued reference to FIG. 10, if perception system 402 determines that no additional orientation interval remains to be processed to determine its local minimum of $f(\theta)$ ("No" at block 1010), process 1000 ends.

[0113] Sensors that generate data points 604 may inadvertently detect other objects, or may generate inaccurate or otherwise unwanted data. Embodiments of the disclosure may thus also provide methods and processes for rejection of outlying data points. FIG. 11 illustrates one such exemplary process. Any manner of outlier rejection is contemplated.

plated. In some embodiments, data point sets **1100** and **1102** may be employed to determine convex hulls, tangents thereto, bounding box orientation intervals, and optimized orientations as described herein. Additionally, outlying data point sets **1104**, **1106** that do not contribute to the cost function $f(\theta)$, e.g., do not change a determined minimum distance, or that only contribute below some threshold, may be rejected prior to determination of convex hull tangents. For example, discrete or isolated sets of points **1104**, **1106** may be identified as being spatially separate from point sets **1100**, **1102**, and their cost functions may be determined. When such cost functions fall below a predetermined threshold value, such as the case of sufficiently small point clusters, the point sets **1104**, **1106** may be neglected in the bounding box calculation processes described herein. In this manner, points that may be erroneously considered as corresponding to a vehicle, such as vehicle exhaust clouds and the like, may be disregarded in determining proper bounding boxes.

[0114] Various additional example embodiments of the disclosure can be described by the following clauses:

[0115] Clause 1. A device comprising:

[0116] at least one processor; and

[0117] at least one memory storing instructions which, when executed by the at least one processor, cause the at least one processor to:

[0118] generate a bounding box enclosing a cluster of points corresponding to at least one portion of a detected vehicle, the bounding box comprising a first side and a second side;

[0119] sort the points into a first set of points located within a distance from the first side and a second set of points located within the distance from the second side;

[0120] generate a first convex hull based on the first set of points and a second convex hull based on the second set of points;

[0121] determine a plurality of intervals of orientations associated with the bounding box within which the first and second sets of points remain constant based on a portion of the first convex hull and a portion of the second convex hull;

[0122] determine an orientation of the bounding box based on the plurality of intervals of orientations that are associated with the bounding box, the orientation of the bounding box minimizing distances from each point of the first set of points to the first side and distances from each point of the second set of points to the second side; and

[0123] provide data associated with the orientation of the bounding box to control circuitry of an autonomous vehicle, the data associated with the orientation configured to cause the control circuitry to control operation of the autonomous vehicle.

[0124] Clause 2. The device of Clause 1, wherein the instructions, when executed by the at least one processor, further cause the at least one processor to:

[0125] determine the plurality of intervals of orientations associated with the bounding box within which the first and second sets of points remain constant based on determining transition orientations associated with the bounding box, the transition orientations corresponding to transitions of at least one of the points from

a state nearer the first side of the bounding box to a state nearer the second side of the bounding box.

[0126] Clause 3. The device of Clause 2, wherein the instructions, when executed by the at least one processor, further cause the at least one processor to:

[0127] determine the transition orientations based on determining orientations of lines tangent to at least one portion of the first convex hull or the second convex hull.

[0128] Clause 4. The device of Clause 2 or Clause 3, wherein the instructions, when executed by the at least one processor, further cause the at least one processor to:

[0129] for each of the plurality of intervals of orientations associated with the bounding box, re-sort the points into a re-sorted first set of points and a re-sorted second set of points, thereby generating re-sorted sets of points corresponding to the intervals of orientations associated with the bounding box.

[0130] Clause 5. The device of Clause 4, wherein the instructions, when executed by the at least one processor, further cause the at least one processor to:

[0131] for each of the plurality of intervals of orientations associated with the bounding box, generate at least one continuous function based on the corresponding re-sorted set of points.

[0132] Clause 6. The device of Clause 4 or Clause 5, wherein the instructions, when executed by the at least one processor, further cause the at least one processor to:

[0133] for each of the plurality of intervals of orientations associated with the bounding box, determine a locally minimizing orientation associated with the bounding box, the locally minimizing orientation associated with the bounding box locally minimizing, within the interval of orientations associated with the bounding box, distances from each point of the re-sorted first set of points to the first side and distances from each point of the re-sorted second set of points to the second side.

[0134] Clause 7. The device of Clause 6, wherein the instructions, when executed by the at least one processor, further cause the at least one processor to:

[0135] determine the orientation of the bounding box based on the plurality of intervals of orientations that are associated with the bounding box, the orientation of the bounding box minimizing distances from each point of the first set of points to the first side and distances from each point of the second set of points to the second side based on selecting, from among the locally minimizing orientations of the bounding box, an orientation corresponding to the smallest distances.

[0136] Clause 8. The device of any one of Clauses 2 through 7, wherein the instructions, when executed by the at least one processor, further cause the at least one processor to:

[0137] store in the at least one memory at least one of the plurality of intervals of orientations or the transition orientations; and

[0138] determine the orientation of the bounding box based on the plurality of intervals of orientations that are associated with the bounding box, the orientation of the bounding box minimizing distances from each point of the first set of points to the first side and distances from each point of the second set of points to

the second side based on accessing the stored at least one of the intervals of orientations or the transition orientations.

[0139] Clause 9. The device of any one of Clauses 1 through 8, wherein the cluster of points corresponding to at least one portion of the detected vehicle comprises at least one of Light Detection And Ranging (LiDAR) measurements, Radio Detection and Ranging (Radar) measurements, location values determined from at least one image, or ultrasound measurements.

[0140] Clause 10. The device of any one of Clauses 1 through 9, wherein the cluster of points corresponding to at least one portion of the detected vehicle comprises two-dimensional coordinates relative to a reference point associated with the autonomous vehicle, and wherein the determined orientation of the bounding box comprises an angle relative to a coordinate frame associated with the autonomous vehicle.

[0141] Clause 11. A method comprising:

[0142] receiving, with at least one processor, a plurality of location values of detected portions of an object;

[0143] determining, with the at least one processor, a first convex hull of a first set of the location values nearest a first side of a bounding enclosure, and a second convex hull of a second set of the location values nearest a second side of the bounding enclosure;

[0144] determining, with the at least one processor, based on at least a portion of the first and second convex hulls, intervals of orientations of the bounding enclosure within which each location value remains nearest a same one of the sides of the bounding enclosure;

[0145] computing, with the at least one processor, from the intervals, an orientation of the bounding enclosure corresponding to a minimum distance between the edges of the bounding enclosure and the location values; and outputting, with the at least one processor, the determined orientation of the bounding box to control circuitry of a vehicle.

[0146] Clause 12. The method of Clause 11, further comprising determining the intervals of orientations according to orientations of the bounding enclosure that correspond to a transition of one location value from a state nearest one edge of the bounding enclosure to a state nearest another edge of the bounding enclosure.

[0147] Clause 13. The method of Clause 12, wherein the computing further comprises:

[0148] selecting, from within each interval of orientations, that orientation corresponding to the minimum distance between the edges of the corresponding bounding enclosure and the nearest location values, so as to form a set of minimum distance orientations; and

[0149] selecting, from among the set of minimum distance orientations, the orientation corresponding to the smallest minimum distance.

[0150] Clause 14. The method of Clause 13, wherein the selecting those orientations corresponding to the minimum distances further comprises selecting those orientations according to at least one continuous function.

[0151] Clause 15. The method of any one of Clauses 12 through 14, further comprising storing the intervals of orientations in at least one of the memories.

[0152] Clause 16. The method of any one of Clauses 11 through 15, further comprising determining the intervals of

orientations at least in part according to ones of the orientations having sides tangent to pairs of the convex hulls.

[0153] Clause 17. The method of any one of Clauses 11 through 16, wherein the location values comprise at least one of Light Detection And Ranging (LiDAR) measurements, Radio Detection and Ranging (Radar) measurements, location values determined from at least one image, or ultrasound measurements.

[0154] Clause 18. The method of any one of Clauses 11 through 17, wherein the object comprises at least one of a vehicle, a pedestrian, a portion of a roadway, or a roadside object.

[0155] Clause 19. The method of any one of Clauses 11 through 18, further comprising rejecting ones of the location values that do not change the minimum distance, and determining the intervals of orientations of the bounding enclosure at least in part according to ones of the location values that were not rejected.

[0156] Clause 20. The method of any one of Clauses 11 through 19, wherein the bounding enclosure is a bounding box.

[0157] In the foregoing description, aspects and embodiments of the present disclosure have been described with reference to numerous specific details that can vary from implementation to implementation. Accordingly, the description and drawings are to be regarded in an illustrative rather than a restrictive sense. The sole and exclusive indicator of the scope of the invention, and what is intended by the applicants to be the scope of the invention, is the literal and equivalent scope of the set of claims that issue from this application, in the specific form in which such claims issue, including any subsequent correction. Any definitions expressly set forth herein for terms contained in such claims shall govern the meaning of such terms as used in the claims. In addition, when we use the term “further comprising,” in the foregoing description or following claims, what follows this phrase can be an additional step or entity, or a sub-step/sub-entity of a previously-recited step or entity.

1. (canceled)

2. A method comprising:

identifying, using at least one processor, a first set of points associated with an object;

identifying, using the at least one processor, a second set of points associated with the object;

determining, using the at least one processor, an orientation of a bounding enclosure associated with the object based at least in part on a first distance between a first side of the bounding enclosure and the first set of points and a second distance between a second side of the bounding enclosure and the second set of points; and

causing, using the at least one processor, navigation by a vehicle based at least in part on the orientation of the bounding enclosure.

3. The method of claim 2, further comprising:

sorting a point into the first set of points based on a third distance between the point and the first side of the bounding enclosure being less than a fourth distance between the point and the second side of the bounding enclosure.

4. The method of claim 2, wherein identifying the first set of points comprises:

identifying the first set of points, wherein the first set of points are located closer to the first side of the bounding

- enclosure than the second side of the bounding enclosure, wherein identifying the second set of points comprises:
- identifying the second set of points, wherein the second set of points are located closer to the second side of the bounding enclosure than the first side of the bounding enclosure.
5. The method of claim 2, further comprising: receiving sensor data associated with at least one image sensor,
- wherein identifying the first set of points comprises: identifying the first set of points based at least in part on the sensor data, and
- wherein identifying the second set of points comprises: identifying the second set of points based at least in part on the sensor data.
6. The method of claim 5, wherein receiving the sensor data comprises:
- receiving one or more of lidar data, radar data, image data, or ultrasound data associated with the at least one image sensor.
7. The method of claim 2, wherein identifying the first set of points comprises:
- identifying the first set of points, wherein the object comprises one or more of a second vehicle, a pedestrian, a cyclist, debris, or at least a portion of a road.
8. The method of claim 2, wherein identifying the first set of points comprises:
- identifying the first set of points, wherein the first set of points corresponds to a set of two-dimensional coordinates relative to a reference point associated with the vehicle.
9. The method of claim 2, further comprising: identifying a first convex hull of the first set of points; and identifying a second convex hull of the second set of points.
10. The method of claim 2, further comprising: generating the bounding enclosure based on the first set of points and the second set of points.
11. The method of claim 2, wherein determining the orientation of the bounding enclosure comprises:
- determining the orientation of the bounding enclosure, wherein the orientation of the bounding enclosure corresponds to an angle relative to a coordinate frame associated with the vehicle.
12. The method of claim 2, wherein determining the orientation of the bounding enclosure comprises:
- determining the orientation of the bounding enclosure from a plurality of orientations of the bounding enclosure.
13. The method of claim 2, wherein determining the orientation of the bounding enclosure comprises:
- determining the orientation of the bounding enclosure from a plurality of orientations of the bounding enclosure, wherein each orientation of the bounding enclosure of the plurality of orientations of the bounding enclosure is associated with a respective distance between the first side of the bounding enclosure and the first set of points and a respective distance between the second side of the bounding enclosure and the second set of points.
14. The method of claim 2, wherein determining the orientation of the bounding enclosure comprises:

- determining the orientation of the bounding enclosure from a first orientation of the bounding enclosure and a second orientation of the bounding enclosure, wherein the first orientation of the bounding enclosure is associated with a third distance between the first side of the bounding enclosure and the first set of points and a fourth distance between the second side of the bounding enclosure and the second set of points, wherein the second orientation of the bounding enclosure is associated with a fifth distance between the first side of the bounding enclosure and the first set of points and a sixth distance between the second side of the bounding enclosure and the second set of points, and wherein the orientation of the bounding enclosure corresponds to the first orientation of the bounding enclosure based on the third distance being less than the fifth distance, and the fourth distance being less than the sixth distance.
15. The method of claim 2, further comprising: providing data associated with the orientation of the bounding enclosure to control circuitry of the vehicle.
16. A system, comprising:
- at least one processor, and
- at least one non-transitory storage media storing instructions, wherein execution of the instructions by the at least one processor cause the at least one processor to:
- identify a first set of points associated with an object;
- identify a second set of points associated with the object;
- determine an orientation of a bounding enclosure associated with the object based at least in part on a first distance between a first side of the bounding enclosure and the first set of points and a second distance between a second side of the bounding enclosure and the second set of points; and
- cause navigation by a vehicle based at least in part on the orientation of the bounding enclosure.
17. The system of claim 16, wherein the execution of the instructions by at least one processor further causes the at least one processor to:
- sort a point into the first set of points based on a third distance between the point and the first side of the bounding enclosure being less than a fourth distance between the point and the second side of the bounding enclosure.
18. The system of claim 16, wherein to determine the orientation of the bounding enclosure, the execution of the instructions by at least one processor further causes the at least one processor to:
- determine the orientation of the bounding enclosure from a plurality of orientations of the bounding enclosure, wherein each orientation of the bounding enclosure of the plurality of orientations of the bounding enclosure is associated with a respective distance between the first side of the bounding enclosure and the first set of points and a respective distance between the second side of the bounding enclosure and the second set of points.
19. At least one non-transitory storage media storing instructions, wherein execution of the instructions by at least one processor cause the at least one processor to:
- identify a first set of points associated with an object;
- identify a second set of points associated with the object;
- determine an orientation of a bounding enclosure associated with the object based at least in part on a first

distance between a first side of the bounding enclosure and the first set of points and a second distance between a second side of the bounding enclosure and the second set of points; and

cause navigation by a vehicle based at least in part on the orientation of the bounding enclosure.

20. The at least one non-transitory storage media of claim **19**, wherein the execution of the instructions by at least one processor further causes the at least one processor to:

sort a point into the first set of points based on a third distance between the point and the first side of the bounding enclosure being less than a fourth distance between the point and the second side of the bounding enclosure.

21. The at least one non-transitory storage media of claim **20**, wherein to determine the orientation of the bounding enclosure, the execution of the instructions by at least one processor further causes the at least one processor to:

determine the orientation of the bounding enclosure from a plurality of orientations of the bounding enclosure, wherein each orientation of the bounding enclosure of the plurality of orientations of the bounding enclosure is associated with a respective distance between the first side of the bounding enclosure and the first set of points and a respective distance between the second side of the bounding enclosure and the second set of points.

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